CURRICULUM VITAE

Konstantin V. Sokolov, Ph.D.

PRESENT TITLE AND AFFILIATION

Primary Appointment

Professor, Department of Imaging Physics, Division of Diagnostic Imaging, The University of Texas MD Anderson Cancer Center, Houston, TX

Dual/Joint/Adjunct Appointment

Adjunct Professor, Department of Biomedical Engineering, The University of Texas, Austin, TX

Adjunct Professor, Department of Bioengineering, Rice University, Houston, TX

CITIZENSHIP

United States

HOME ADDRESS

218 Flamingo Island Drive
Missouri City, TX 77459
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OFFICE ADDRESS

The University of Texas MD Anderson Cancer Center
1881 East Road
Unit Number: 1902
Houston, TX 77054
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EDUCATION

Degree-Granting Education

Moscow Engineering Physics Institute, Moscow, Russia, Diploma Degree, Summa Cum Laude, 1987, Physics

Moscow State University, Moscow, Russia, PHD, 1992, Biophysics

Postgraduate Training

Postgraduate Training, Laboratory of Biomolecular Spectroscopy, Reims University, Reims, France, 1993

CREDENTIALS

Board Certification

N/A

Licensures

Active

N/A

Inactive

N/A

EXPERIENCE/SERVICE

Academic Appointments

Assistant Scientist, Laboratory of Optical Spectroscopy, Shemyakin and Ovchinnikov Institute of Bioorganic Chemistry, Russian Academy of Sciences, Moscow, Russian Federation, 1992-1993

Associate Scientist, Laboratory of Optical Spectroscopy, Shemyakin and Ovchinnikov Institute of Bioorganic Chemistry, Russian Academy of Sciences, Moscow, Russian Federation, 1993-1994

Visiting Scholar, Department of Chemistry, Iowa State University, Ames, IA, 1994-1997

Assistant Scientist, Department of Chemistry, Iowa State University, Ames, IA, 1997-1998

Research Associate, The University of Texas, Austin, TX, 1998-2002

Assistant Professor (Tenure Track), Department of Imaging Physics - Research, Division of Diagnostic Imaging, The University of Texas MD Anderson Cancer Center, Houston, TX, 10/2002-8/2008

Adjunct Assistant Professor, Department of Biomedical Engineering, The University of Texas, Austin, TX, 10/2002-8/2008

Associate Professor, Department of Imaging Physics, Division of Diagnostic Imaging, The University of Texas MD Anderson Cancer Center, Houston, TX, 9/2008-8/2014

Adjunct Associate Professor, Department of Biomedical Engineering, The University of Texas, Austin, TX, 9/2008-8/2014

Adjunct Professor, Department of Biomedical Engineering, The University of Texas, Austin, TX, 9/2014-present

Adjunct Professor, Department of Bioengineering, Rice University, Houston, TX, 9/2014-present

Professor, Department of Imaging Physics, Division of Diagnostic Imaging, The University of Texas MD Anderson Cancer Center, Houston, TX, 9/2014-present

Section Chief Research, Department of Imaging Physics, Division of Diagnostic Imaging, The University of Texas MD Anderson Cancer Center, Houston, TX, 9/2019-8/2022 (it is a 3-year term position)

Administrative Appointments/Responsibilities

Director of Education, Department of Imaging Physics, Division of Diagnostic Imaging, The University of Texas MD Anderson Cancer Center, Houston, TX, 2023-present

Other Appointments/Responsibilities

IGERT faculty, The National Science Foundation Integrative Graduate Education and Research Traineeship Program: Cellular and Molecular Imaging for Diagnostics and Therapeutics, Department of Biomedical Engineering, The University of Texas at Austin, Austin, TX, 2002-2008

Faculty Member, Center for Nano & Molecular Science and Technology, The University of Texas, Austin, TX, 2005-2014

Faculty Member, The UT Graduate School of Biomedical Sciences, Houston, TX, 12/2013-present

Director, T32 Interdisciplinary Translational Pre/Postdoctoral Program in Cancer Nanotechnology, University of Texas MD Anderson Cancer Center, Houston, TX, 9/2015-present

Faculty Member, Image Guided Cancer Therapy Research Program, Division of Diagnostic Imaging, University of Texas MD Anderson Cancer Center, Houston, TX, 2017-present

Endowed Positions

N/A

Consultantships

N/A

Military or Other Governmental Service

N/A

Institutional Committee Activities

Graduate Study Committee, Department of Biomedical Engineering, University of Texas, Austin, Texas, Member, 2002-present

Graduate Fellowship Awards Committee, Department of Biomedical Engineering, University of Texas, Austin, Texas, Member, 2003-2011

Research Strategy Committee, Department of Imaging Physics, University of Texas M D Anderson Cancer Center, Houston, Texas, Member, 2005-2014

Education Program, Department of Biomedical Engineering, University of Texas MD Anderson Cancer Center, Director, 2007-2009

The Alliance for NanoHealth, Houston, TX, Member, 2007-present

GSC Executive Committee, Department of Biomedical Engineering, University of Texas, Austin, Texas, Member, 2009-2011

Seminars Committee, Department of Biomedical Engineering, University of Texas, Austin, Texas, Member, 2010-2011

Mock Review Study Section, Department of Imaging Physics, University of Texas MD Anderson Cancer Center, Chair, 2014-2015

Research Faculty Search Committee, Department of Imaging Physics, University of Texas MD Anderson Cancer Center, Chair, 2014-2016

Mock Study Section, University of Texas MD Anderson Cancer Center, Member, 2014-2016

Research Lab Space Committee, University of Texas MD Anderson Cancer Center, Member, 2014-2017

Institutional Research Grant (IRG) program, University of Texas MD Anderson Cancer Center, Member, 2014-present

Executive Committee, Department of Imaging Physics, Member, 2015-present

University of Texas MD Anderson Pathway in Systems Imaging Steering Committee, Member, 2016-2019

INTEREST Review Study Section, University of Texas MD Anderson Cancer Center, Member, 2016

Promotion and Tenure Committee, Division of Diagnostic Imaging, University of Texas MD Anderson Cancer Center, Member, 2017-present

Pediatrics Research Imaging Team, Division of Pediatrics, University of Texas MD Anderson Cancer Center, Member, 2018-present

Governance Committee, Department of Imaging Physics, Member, 2020-present

Technology Task Force Committee, University of Texas MD Anderson Cancer Center, Member, 2020-present

Education Committee for Health Professions, University of Texas MD Anderson Cancer Center, Member, 2021-present

Hybrid Residency Admission Committee, Department of Imaging Physics, Member, 2021- 2022

Executive Oversight Committee for the Advanced Microscopy Core, University of Texas MD Anderson Cancer Center, Member, 2022-present

Deputy Chair, Division of Diagnostic Imaging (DI) Research Functional Committee, 2023-present

Director of Education, Department of Imaging Physics, 2023-present

HONORS AND AWARDS

Young Scientist Award, Shemyakin Institute of Bioorganic Chemistry USSR Academy of Sciences, 1989

Fellowship, Centre National de la Recherché Scientifique (CNRS) France, 1992-1993

Outstanding Young Scientist Award, Russian Academy of Sciences, 1993

Whitaker Foundation Award, 2000

Chair, The Molecular Probes and Nanobio-optics Technical Group of the Optical Society of America (OSA), 2008-2011

Speaker, Lester Wolfe Workshop in Laser Biomedicine, MIT and Harvard-MIT Division of Health Science and Technology, 2008

Chair, OSA Topical Meeting: Optical Molecular Probes, Imaging, and Drug Delivery, 2011

Chair, III International Symposium on Topical Problems of Biophotonics, 2011

Plenary Speaker, XVI International School on Optics, Laser Physics and Biophotonics, Saratov, Russia, 2012

General Chair, OSA Topical Meeting: Optical Molecular Probes, Imaging, and Drug Delivery, 2013

Faculty Leadership Academy, M.D. Anderson Cancer Center, 2014-2015

Chair, V International Symposium on Topical Problems of Biophotonics, Nizhny Novgorod, Russia, 2015

Elected Fellow, Optical Society of America, 2015

Senior Member, The International Society for Optics and Photonics (SPIE), 2017

Elected Fellow, The International Society for Optics and Photonics (SPIE), 2018

Distinguished Investigator, Council of the Academy of Radiology and Biomedical Imaging Research, 2020

Elected Fellow, The American Institute for Medical and Biological Engineering (AIMBE), 2023

RESEARCH

Grants and Contracts

Funded

Principal Investigator, 15%, In situ cancer cell specific biomineralization to overcome nanoparticle delivery barriers and sensitize pancreatic cancer to radiotherapy, R01CA274415-01, NIH/NCI, MPI – Sokolov (contact), Krishnan and Cho, 8/1/2022-7/31/2027, $3,001,512 ($600,302/year)

Principal Investigator, 10%, Interdisciplinary Translational Pre/Postdoctoral Program in Cancer Nanotechnology, 2T32CA196561-06, NIH/NCI, 4/1/2022-3/31/2027, $2,695,935 ($539,187/year)

Principal Investigator, 5%, Hyperpolarized ^{129}Xe MRI for imaging NK cell therapy of lung metastasis, 1R21CA280799-01, NIH/NCI, MPI – Sokolov (contact) and Bankson, 5/1/2023-4/30/2025, $275,000 ($150,000/year)

Principal Investigator, 5%, Radiosensitization of thyroid cancer by cancer cell specific reduction of gold ions, 1R21CA259839-01, NIH/NCI, MPI – Sokolov (contact) and Lai, 2/19/2022-1/31/2025 (NCE), $416,543 ($243,000/year)

Co-Investigator, 15%, Molecular Photoacoustic Imaging for Diagnostics and Therapy Monitoring, 1R01EB028762-01, NIH/NIBIB, PI - Bouchard, Richard, 8/1/2020-4/30/2025, $3,068,180 ($613,636/year)

Co-Investigator, 10%, Treatment planning of ADC therapy for ovarian cancer with molecular photoacoustic-ultrasonic imaging, RP240311, CPRIT, PI - Bouchard, Richard, 3/01/2024-2/28/2027, $997,368 ($332,456/year)

Co-Investigator, 2%, The Center for Innovation and Translation of Point of Care Technologies for Equitable Cancer Care (CITEC), U54, NIH/NCI, PI – Richards-Kortum, 9/1/2023-7/30/2028, $42,240.

Co-Investigator, 5%, A Low-Cost, Rugged, End-to-End System for Slide-Free Pathology Based on Computational Microscopy and Pan-Cancer Targeted Vital Dyes, PSI-027, ARPA-H, PI-Richards-Kortum, 06/01/2024 – 05/31/2029, $242,205 ($48,441/year).

Pending

Co-Investigator, 10%, Novel approach to NK cell therapy for recurrent/refractory pediatric medulloblastoma, 1UG3CA283650-01, NIH/NCI, PI – Wintermark/Gopalakrishnan, 12/01/2024 – 11/30/2029, $640,728 ($213,576/year).

Principal Investigator, 5%, Quantitative noncontact imaging of cellular mechanotransduction, 1R21CA297102-01, NIH/NCI, 12/1/2024-11/3/2026, $445,500 ($150,000/year).

Co-Investigator/Project 2 Co-Lead, 10%, New approach for osteosarcoma: Enhancing and monitoring NK cell therapy, 1P01CA275719-01A1, NIH/NCI, PI – Gopalakrishnan, 7/01/2024-6/30/2029, $2,703,685 ($540,737/year).

Principal Investigator, 5%, Phase-change nanodroplets for ultrasound guided cancer immunotherapy, 1R21CA290267-01A1, 12/01/2024-11/30/2026, $445,500 ($150,000/year).

Project Leader, 15%, Using stealth gold nanoparticles to improve radiation therapy, RP240608, CPRIT MIRA, PI-Krishnan, 08/31/2024 – 08/31/2028, $947,896 ($236,974/year).

Other

NA

Completed

Principal Investigator, 50%, Polarization Spectroscopy to Measure Nuclear Size Distribution: Improved Pre-Cancer Detection, Whitaker Foundation, 9/1/2000-8/31/2003, $239,545 ($79,973/year)

Co-Principal Investigator, 25%, Biospecific Contrast Agents for Pre-Cancer Detection, BES-0119450, National Science Foundation (NSF), PI - Richards-Kortum, 9/1/2001-8/31/2004, $674,689 ($224,896/year)

Co-Principal Investigator, 15%, In Vivo Imaging of Human Papillomavirus Infection, NIH/NCI, PI - Richards-Kortum, 1/1/2003-12/31/2006, $1,370,575

Principal Investigator, 20%, Polarization Imaging of Pre-Cancer, TF-03-0013, Whitaker Foundation, 2/1/2004-1/31/2005, $65,063

Principal Investigator-MDACC, 20%, MDACC, Optical Systems for In Vivo Molecular Imaging of Cancer, RO1 CA103830-06 BRP, NIH/NCI, PI - Richards-Kortum, 8/1/2004-5/31/2009, $6,970,060

Principal Investigator, 20%, Polarization Spectroscopy and Imaging of Pre-Cancer, R01 EB003540, NIH/NIBIB, 5/11/2005-2/28/2011, $720,000

Principal Investigator, Multifunctional Nanoparticles for MRI and Optical Imaging, The University of Texas SPORE in Lung Cancer, 12/1/2005-4/15/2006, $24,860

Mentor, 5%, Medical Physics Training Grant for Image-Guided Therapy Research, T32 CA119930, NIH/NCI, PI - Jackson, 8/4/2006-7/31/2011

Co-Principal Investigator, 2%, Early Cancer Detections with Two Photon Luninescence Gold Nanorods, NIH/NCI, PI - Ben-Yakar, 9/1/2006-8/31/2008, $150,000

Principal Investigator, Molecular Targeted Stellated Nanoparticles for Second Harmonic Generation Imaging of Cancer, Tate Foundation Developmental Seed grant, 2/1/2007-1/31/2008, $20,000

Principal Investigator, 12%, Acoustic Imaging of Sentinel Node Metastasis Using Plasmonic Nanosensors, R01 EB008101, NIH/NIBIB, 9/1/2007-8/31/2011, $900,000 (total direct costs)

Principal Investigator, 8%, Aptamer-siRNA Chimera/Nanoparticle Conjugates for MRI Guided Cancer Therapy, 1-R21 CA135315-01, NIH/NCI, 6/1/2008-6/30/2012, $250,000

Co-Investigator, 8%, Magneto-Motive Ultrasound Imaging Using Molecular Specific Nanoparticles, R01 EB 008821, NIH/NIBIB, PI - Emelianov, 6/1/2008-3/31/2013, $1,250,000

Principal Investigator, Aptamer-siRNA Chimera/Nanoparticle Conjugates for MRI Guided Cancer Therapy, ARRA 3 R21 CA135315-02S1, NIH/NCI, 7/1/2009-5/31/2010, $116,934

Mentor, Comprehensive Training Program Imaging Science and Informatics, NIH/NIBIB, PI - Rylander, G, Markey M, 8/1/2009-7/31/2014

Co-Investigator, 9%, Combined Intravascular Ultrasound and Photoacoustic Imaging of Atherosclerosis, ARRA R01 HL096981, NIH/NHLBI, PI - Emelianov, 9/1/2009-8/31/2011, $999,864

Principal Investigator, 8%, Biodegradable Plasmonic Nanoparticles for Cancer Imaging and Therapy, R01 CA143663, NIH/NCI, 1/1/2010-12/31/2015, $1,367,061 ($191,437/year)

Co-Investigator, 8%, Molecular Photothermal Therapy of Cancer Using Targeted Metal Nanoparticles, R01 CA 149740, NIH/NCI, PI - Emelianov, 4/1/2010-3/31/2015, $1,000,000 ($25,000/year)

Co-Investigator, 4%, Electrically-connected plasmonic metamaterials for capture and detection of CTC, R21-CA 173663, NIH/NCI, PI - Shvets, 6/1/2013-5/31/2015, $275,000 ($150,000/year)

Principal Investigator, iThera Medical Multispectral Optoacoustic Tomography System for Full Body Molecular and Functional Small Animal Imaging, 1S10OD019946-01, NIH/NCI, 4/1/2015-3/31/2016, $501,855

Principal Investigator, Multimodal nanoparticles for adoptive immunotherapy monitoring, MD Anderson Cancer Center, 9/1/2015-8/31/2016, $75,000

Mentor, Re-activating Anti-Tumor Immunity Using Non-Coding RNAs, 373307, Ovarian Cancer Research Fund, PI - Wu, Sherry, 2/1/2016-1/31/2019, $450,000 ($150,000/year)

Mentor, Magnetic relaxometry detection of stealth, antibody-targeted micellar iron oxide nanoparticles in-vivo, American Association for Cancer Research (AACR), PI - Romero Aburto, Rebeca, 1/1/2017-4/5/2017 ($1,500/year)

Principal Investigator-MDACC, 4%, Optical Systems for In Vivo Molecular Imaging of Cancer, 1R01CA103830-06 BRP, NIH/NCI, PI - Richards- Kortum, 9/23/2011-7/31/2018, $3,400,000 ($75,304/year)

Principal Investigator, 1%, Development of unconventional MRI approaches to imaging cancer immunotherapy, CABI MI2, GE Healthcare, 1/1/2016-6/30/2018 ($80,000/year)

Co-Mentor, Simultaneous Thermal and Osmotic Stresses in Tumor Ablation: Imaging and Biology, 3R01CA201127-02S1, NIH/NCI, PI - Cressman, Erik, 8/1/2017-6/30/2019, $509,332 ($127,333/year)

Collaborator, 5%, MRI imaging of adoptive NK Cell immunotherapy, IRG-Institutional Research Grant, MDACC, PI - Gopalakrishnan, 9/1/2018-8/31/2020, $75,000/year

Principal Investigator, 12%, Acoustic Imaging of Sentinel Node Metastasis Using Plasmonic Nanosensors, R01 EB008101, NIH/NIBIB, 4/1/2012-2/28/2019, $1,980,139

Project Leader, 9%, Advancing magnetic relaxometry, Senior Scientific, PI - Hazle, 11/1/2015-8/31/2020, $1,755,837 ($179,600/year)

Principal Investigator, 9%, Functional, cellular and molecular imaging using using ultrasound-guided photoacoustics, R01 CA158598, NIH/NCI, MPI – Emelianov, Homan and Sokolov, 10/1/2012-6/30/2020, $2,500,000 ($153,292/year)

Principal Investigator, 5%, Biodegradable Nanoclusters for Molecular Cancer Imaging, RP170314, Cancer Prevention & Research Institute of Texas (CPRIT), 12/1/2016-11/30/2020, $703,592 ($284,951/year)

Co-Investigator, 5%, Assessments of Tumor Perfusion with Dynamic Contrast Enhanced Multispectral Optoacoustic Tomography, RP190211, Cancer Prevention & Research Institute of Texas (CPRIT), PI - Pagel, Mark, 3/1/2019-10/31/2020 ($278,772/year)

Co-Investigator, 5%, Exploiting Melanin Synthesis to Improve Tumor Detection with Multispectral Optoacoustic Tomography and Optical Guided Surgery, 1R21EB028062-01A1, NIH/NIBIB, PI - Pagel, Mark, 4/1/2020-10/31/2020, $275,000 ($150,000/year)

Principal Investigator, 10%, Interdisciplinary Translational Pre/Postdoctoral Program in Cancer Nanotechnology, 5T32CA196561-05, NIH/NCI, 9/1/2015-8/31/2021, $1,830,024, ($211,206/year)

Co-Investigator, 4%, Targeted Phase-change Nanodroplets for Translatable Molecular Ultrasound Imaging of Micrometastases, R21CA234526-02, NIH/NCI, PI - Bouchard, Richard, 12/1/2018-11/30/2021, NCE, $275,000 ($150,000/year)

Principal Investigator, 4%, 3D Imaging of Tumor Mechanobiology Using Nanobomb Optical Coherence Elastography, R21CA231561-02, NIH/NCI, 2/1/2019-1/31/2022, NCE, $275,000 ($150,000/year)

Co-Leader – P2, 5%, CAR-NK Adoptive Cell Therapy for Osteosarcoma, Multidisciplinary Research Program (MRP), MDACC, PI - MPI: Kundra, Vikas and Gopalakrishnan, Vidya, 9/1/2019-8/31/2022, $250,000 ($100,000/year)

Principal Investigator, 5%, In situ cancer cell specific synthesis of gold nanoclusters for radiosensitization of pancreatic cancer, 1R21CA252156-01, NIH/NCI, MPI – Sokolov (contact) and Krishnan, 7/1/2020-12/31/2022, NCE, $275,000 ($150,000/year)

Principal Investigator, 5%, In situ cancer cell specific synthesis of gold nanoclusters for enhanced radiotherapy of head and neck cancer, IRG, MDACC, 6/1/2021-5/31/2023, $75,000

Co-Investigator, 10%, Neoadjuvant treatment response monitoring of breast cancer with molecular photoacoustic imaging, RP190131, Cancer Prevention & Research Institute of Texas (CPRIT), PI - Bouchard, Richard, 3/1/2019-2/29/2024 (NCE), $895,908 ($298,636/year)

Principal Investigator, 10%, High sensitivity 19F MRI for clinically translatable imaging of adoptive NK cell brain tumor therapy, RP200223, Cancer Prevention & Research Institute of Texas (CPRIT), MPI – Sokolov (contact) and Bankson, 3/1/2020-2/29/2024 (NCE), $887,713 ($295,904/year)

Principal Investigator, 6%, Development of fluorinated dyes for deeper tissue photoacoustic imaging with phase changing nanodroplets, 1R21EB030805-01A1, NIH/NIBIB, 7/1/2021-4/30/2024 (NCE), $445,500 ($243,000/year)

Not Funded

Principal Investigator, 20%, In Vivo Molecular Imaging of Sentinel Node Metastasis, 1 R21 CA 110099-01, NIH/NIBIB, 7/1/2005-6/30/2009, $1,524,611

Co-Investigator, 8%, Optical Spectral Tomography with Molecular Specific Nanorod Markers, RO1, NIH/NCI, PI - Perelman, 6/1/2010-5/31/2015, $1,250,000

Principal Investigator, 12%, Synergism of Nanoparticles with Molecular Ligands for Cancer Imaging and Therapy, R01 CA151995-01, NIH/NCI, 7/1/2010-6/30/2015, $2,160,511

Co-Investigator, 8%, Spectro-Polarimetric Microscope for multiplexed imaging with plasmonic particles, R21 EB013397-01, NIH/NIBIB, 1/1/2011-12/31/2012, $275,000

Principal Investigator, 6%, SERS probes for imaging of molecular interactions, R21 EB016367-01, NIH/NIBIB, 9/1/2012-8/31/2014, $275,000

Principal Investigator, 51%, Light switchable “back-packs" for cells: applications in immunotherapy, DP1 OD009521-01, NIH/NCI, 9/30/2012-7/31/2017, $2,500,000

Principal Investigator, 8%, Multimodal Intra-operative Assessment of Solid Tumor Resection Margins, R01 CA172476-01, NIH/NCI, 10/1/2012-9/30/2017, $2,225,587

Principal Investigator, 12%, Blueprint for Molecular Nanotherapeutics: Targeting Receptors in Breast Cancer, R01, NIH/NIBIB, 4/1/2013-3/31/2018, $2,465,959

Co-Investigator, 8%, Dual ultrasound/photoacoustic contrast for molecular imaging of breast cancer, R01, NIH/NCI, PI - Esteva, 4/1/2013-3/31/2018, $2,221,511

Principal Investigator, 8%, Synergistic Photoacoustic Activation and Monitoring of shRNA Delivery, R01EB017471-01, NIH/NIBIB, 4/1/2013-3/31/2018, $2,390,610

Principal Investigator, 4%, Label-Free Multimodal Assessment of Tissue Architecture, Morphology, R21GM107934-01, NIH/NIGMS, 7/1/2013-6/30/2016, $375,000

Principal Investigator, 9%, Superparamagnetic nanoclusters for immunotherapy guidance and monitoring, RP150417, Cancer Prevention & Research Institute of Texas (CPRIT), 3/1/2015-2/28/2018, $854,148 ($282,376/year)

Principal Investigator, 6%, Magneto-plasmonic nanoparticles for adoptive immunotherapy guidance and monitoring, 1R21CA195115-01, NIH/NCI, 3/31/2015-3/30/2017, $275,000

Principal Investigator, 4%, Versatile approach for capture and characterization of circulating tumor cells, R21CA187662, NIH/NCI, 7/1/2015-6/30/2016, $275,000 ($150,000/year)

Principal Investigator, 17%, Biodegradable Plasmonic Nanoparticles for Cancer Imaging and Therapy, 2R01CA143663-06, NIH/NCI, 7/1/2015-6/30/2020 ($493,362/year)

Principal Investigator, 6%, Multimodal nanoagent for guidance and imaging of cancer immunotheraphy, 1R21CA201801-01, NIH/NCI, 8/31/2015-8/30/2017, $275,000

Co-Investigator, 4%, Nanotechnology solutions for stromal and immunological barriers in cancer, U54 Center of Cancer Nanotechnology Excellence (CCNE), NIH/NCI, PI - Anil Sood, 9/1/2015-8/31/2020, $55,816

Co-Investigator, 10%, Targeting ovarian cancer microenviroment using exosomal therapeutics, RP16097, Cancer Prevention & Research Institute of Texas (CPRIT), PI - Sood, 2/1/2016-1/31/2019, $119,864

Co-Investigator, 15%, MR Imaging of Natural Killer Cell Immunotheraphy, Cancer Prevention & Research Institute of Texas (CPRIT), PI - Schellingerhout, 9/1/2016-8/31/2021 ($247,200/year)

Principal Investigator, 17%, Hydrophobic shielding to enhance nanoparticle mediated radiosensitization of tumors, 1U01 CA 203319-01A, NIH/NCI, 11/30/2016-11/29/2021, $2,226,554

Co-Principal Investigator, 5%, Superparamagnetic Relaxometry for Early Detection of Ovarian Cancer, Sister Institution Network Fund, PI - D. Fuentes, 6/2017-5/2019, $100,000 ($50,000/year)

Co-Investigator, 5%, A data driven superparamagnetic relaxometry signal model for ovarian cancer detection in mouse models, 1R21CA222999-01, NIH/NCI, PI - Fuentes, David and Kelsey Mathieu, 12/1/2017-11/30/2019, $275,000 ($150,000/year)

Principal Investigator, Multiphoton imaging system Leica SP8 for functional imaging of small cancer models, 1S10OD025023-01, NIH, 2/1/2018-1/31/2019, $1,065,745 ($1,065,745/year)

Co-Investigator, Molecular photoacoustic imaging for cancer diagnostics and therapy monitoring, 1 R01 CA217896-01A1, NIH/NCI, PI - Sood, Anil and Harris, Justin Thomas, 4/1/2018-3/1/2023 ($513,383/year)

Advisory Committee (Admin Core), Accelerating the clinical advancement of gold nanopoarticles as radiation amplifying agents, 1P01CA225521-01, NIH/NCI, PI - Krishnan, Sunil; Schuemann, Jan Patrick Oscar, 4/1/2018-3/31/2023, $9,550,900 ($1,900,333/year)

Co-Investigator, 3%, Molecular photoacoustic imaging using liposome-encapsulated dye J-aggregates for predicting response to neoadjuvant breast cancer therapy, 1 R21 CA231607-01, NIH/NCI, PI - Bouchard, Richard, 7/1/2018-6/30/2020, $281,238 ($127,492/year)

Principal Investigator, 4%, 3D Imaging of Tumor Mechanobiology Using Nanobomb Optical Coherence Elastography, 1R21CA231561, NIH/NCI, 7/1/2018-8/31/2020, $275,000 ($125,000/year)

Co-Principal Investigator, 3%, Molecular Ultrasound Imaging using Targeted Phase-change Nanodroplets for Accurate Assessment of Sentinel Lymph Node Micrometastases, RP180780, Cancer Prevention & Research Institute of Texas (CPRIT), PI - Bouchard, Richard, 8/31/2018-8/30/2020 ($94,547/year)

PI, Project 1, 15%, Imaging of Natural Killer Cell Immunotherapy, RP180777, Cancer Prevention & Research Institute of Texas (CPRIT), PI - Schellingerhout, Dawid, 8/31/2018-8/30/2022, $5,694,380 ($1,365,950/year)

Co-Investigator, 5%, A data driven superparamagnetic relaxometry signal model for ovarian cancer detection, None, National Science Foundation (NSF), PI - Fuentes, David and Mathieu, Kelsey, 9/1/2018-8/31/2022, $875,918 ($226,780/year)

Co-Investigator, 12%, Molecular Photoacoustic Imaging for Cancer Diagnostics and Therapy Monitoring, 1R01CA235563-01, NIH/NCI, PI - Bouchard, Richard, 9/1/2018-8/31/2023 ($471,362/year)

Principal Investigator, 4%, Monitoring treatment response of neoadjuvant breast cancer therapy with molecular photoacoustic imaging, BC171686, Department of Defense (DOD), 10/1/2018-9/30/2021, $208,018

Co-Investigator, 5%, Exploiting Melanin Synthesis to Improve Tumor Detection during Surgery, Mark Foundation/ASPIRE, PI - Pagel, Mark, 12/1/2018-1/30/2019, $242,596 ($242,596/year)

Principal Investigator, 4%, Advancing 19F MRI NK cell tracking probes using amphiphilic peptide platform, 1R21CA235593-01, NIH/NCI, 12/1/2018-11/30/2020 ($275,000/year)

Co-Investigator, 5%, Assessments of Tumor Perfusion with Dynamic Contrast Enhanced Multispectral Optoacoustic Tomography, 1R01CA233637-01, NIH/NCI, PI - Pagel, Mark, 12/1/2018-11/30/2023 ($486,854/year)

Co-Investigator, 5%, Superparamagnetic Relaxometry Detection of Ovarian Cancer, 1R01EB027477-01, NIH/NIBIB, PI - Fuentes, David and Mathieu, Kelsey, 12/1/2018-11/30/2023, $2,721,051 ($559,223/year)

Co-Investigator, 10%, Assessing Brain Tumor Biomechanics in Response to Therapy with Optical Coherence Elastography, Cancer Prevention & Research Institute of Texas (CPRIT), PI - Larin, Kirill, 3/1/2019-2/28/2022 ($97,280/year)

Principal Investigator, 15%, Hydrophobic shielding to enhance nanoparticle mediated radiosensitization of tumors, RP190179, Cancer Prevention & Research Institute of Texas (CPRIT), 3/1/2019-2/28/2022 ($284,780/year)

Co-Investigator, 5%, NK Cell Therapy for Refractory or Relapsed Osteosarcoma, MDACC, PI - MPI: Kundra, Gordon, Najjar, Gopalakrishnan, 3/15/2019-3/14/2022, $250,000 ($250,000/year)

Co-Investigator, 5%, Exploiting Melanin Synthesis for Reporter Gene Imaging with Multispectral Optoacoustic Tomography, 1R21EB028045, NIH/NIBIB, PI - Pagel, Mark, 4/1/2019-3/31/2021, $275,000

Principal Investigator, 5%, Perfluorocarbon Nanodroplets to Overcome Hypoxia in Tumor Radiosensitization, 1R21CA239186-01, NIH/NCI, 4/1/2019-3/31/2021, $275,000

Core Leader: Core 3, Biodistribution and Toxicity, 15%, Accelerating the clinical advancement of gold nanopoarticles as radiation amplifying agents, 1P01CA225521-01A1, NIH/NCI, PI - Krishnan, Sunil; Schuemann, Jan Patrick Oscar, 4/1/2019-3/31/2024, $673,780 ($149,612/year)

Co-Investigator, 5%, Assessments of Tumor Perfusion with Dynamic Contrast Enhanced Multispectral Optoacoustic Tomography, 1R01CA233637-01A1, NIH/NCI, PI - Pagel, Mark, 4/1/2019-3/31/2024, $2,473,416 ($489,388/year)

Co-Investigator, 12%, Molecular photoacoustic imaging for cancer diagnostics and therapy monitoring, 1R01CA235563-01A1, NIH/NCI, PI - Bouchard, Richard, 4/1/2019-3/31/2024 ($471,362/year)

Co-Investigator, 6%, Molecular photoacoustic-ultrasonic imaging for predicting response to neoadjuvant chemotherapy for triple negative breast cancer, BC181549, DOD/Congressionally Directed Medical Research Programs (DOD/CDMRP), PI - Bouchard, Richard, 5/1/2019-4/30/2022, $749,218 ($246,472/year)

Co-Investigator, 3%, Molecular photoacoustic imaging using liposome-encapsulated dye J-aggregates for predicting response to neoadjuvant breast cancer therapy, R21CA231607-01A1, NIH/NCI, PI - Bouchard, Richard, 7/1/2019-6/30/2021, $275,000 ($125,000/year)

Principal Investigator, 5%, Perfluorocarbon Nanodroplets to Overcome Hypoxia in Tumor Radiosensitization, 1R21CA239186-01A1, NIH/NCI, 7/1/2019-6/30/2021, $275,000 ($125,000/year)

Co-Investigator, 5%, Superparamagnetic relaxometry detection of metastatic breast cancer, 1R21CA240572-01, NIH/NCI, PI - Fuentes, Mathieu, 7/1/2019-6/30/2021, $275,000 ($150,000/year)

MPI, 10%, High sensitivity 19F MRI for clinically translatable imaging of adoptive NK cell brain tumor therapy., 1R01CA240421-01, NIH/NCI, PI - Bankson, Sokolov, Gopalakrishnan, 7/1/2019-6/30/2024, $2,641,475 ($541,433/year)

Principal Investigator, 20%, Hydrophobic shielding of nanoparticles to improve circulation time, 1R01CA242987-01, NIH/NCI, 7/1/2019-6/30/2024, $2,244,336 ($449,658/year)

Mentor, Intracellular gold nanoparticles biosynthesis with subsequent exosomal packaging for targeted therapy against invasive ovarian cancer, HERA Women's Cancer Foundation, PI - Schwartz-Duval, Aaron, 9/1/2019-8/31/2020, $19,902 ($19,902/year)

Mentor, Image Guided Cancer Therapy Training Program, 1T32CA244131-01, NIH/NCI, PI - Brock, Kristy, 9/1/2019-8/31/2024, $1,245,324 ($248,460/year)

Collaborator, 5%, Superparamagnetic relaxometry detection of lymph node involvement in HER2+ breast cancer, RP200353, Cancer Prevention & Research Institute of Texas (CPRIT), PI - Fuentes, 3/1/2020-2/28/2023, $853,681 ($284,441/year)

Principal Investigator, 5%, Perfluorocarbon Nanodroplets to Overcome Hypoxia in Tumor Radiosensitization, 1R21CA249268-01, NIH/NCI, 4/1/2020-3/31/2022, $275,000 ($125,000/year)

Co-Investigator, 5%, Tumor heterogeneity in immune sensitivity of DIPGs, Chance of Life Foundation, PI - Gopalakrishnan, 7/1/2020-12/31/2021, $300,000 ($213,345/year)

Co-Investigator, 3%, Direct 3D Printing of Anatomically and Functionally Mimicking Photoacoustic-ultrasound Imaging Phantoms, 1R21CA252926-01, NIH/NCI, PI - Bouchard, Richard, 7/1/2020-6/30/2022, $275,000 ($150,000/year)

Co-Investigator, 5%, Molecular dynamics study of alkylated PEG coating for stealth gold nanoparticles, 1R21GM138868-01, NIH/NCI, PI - Fuentes, David, 7/1/2020-6/30/2022, $275,000 ($125,000/year)

Principal Investigator, 20%, Shielded gold nanoparticles with greatly improved pharmacokinetics for radiosensitization of head and neck tumors, R01 DE030017-01, NIH/NCI, 7/1/2020-6/30/2025, $1,538,159 ($308,800/year)

Principal Investigator, 5%, Development of antibody-conjugated magnetic liposomal nanoparticles for early detection of ovarian cancer by magnetic relaxometry, None, Internal Moonshot, 9/1/2020-8/31/2022, $200,000 ($100,000/year)

Principal Investigator, 5%, Development of fluorinated dyes for deeper tissue photoacoustic imaging with phase changing nanodroplets., 1R21EB030805-01, NIH/NCI, 9/1/2020-8/31/2022, $275,000 ($125,000/year)

Principal Investigator-MDACC, 5%, Radiosensitization of pancreatic cancer with gold laden perfluorocarbon nanodroplets, 1R21CA256667-01, NIH/NCI, PI - Sokolov (contact), Krishnan, 9/1/2020-8/31/2022, $275,000 ($125,000/year)

Principal Investigator-MDACC, 5%, Radiosensitization of pancreatic cancer with gold laden perfluorocarbon nanodroplets, 1R21CA256667-01, NIH/NCI, PI - Sokolov (contact), Krishnan, 9/1/2020-8/31/2022, $275,000 ($125,000/year)

Co-Investigator, 5%, Detection of Protease Activity in Metastatic Breast Cancer with Photoacoustic Imaging, BC190371, DOD/Congressionally Directed Medical Research Programs (DOD/CDMRP), PI - Pagel, Mark, 10/1/2020-9/30/2023, $448,597 ($149,982/year)

Principal Investigator-MDACC, 20%, Shielded gold nanoparticles with greatly improved pharmacokinetics for radiosensitization of head and neck tumors, R01 DE030017-01A1, NIH/NCI, PI - Krishnan (contact), Sokolov, Cho, 12/1/2020-11/30/2025 ($297,110/year)

Principal Investigator, 15%, Stealth nanoscale armored particles for radiosensitization of head and neck tumors, 1R01CA264096-01, NIH/NCI, MPI – Krishnan, Cho, Sokolov, 7/1/2021-6/30/2026, $1,175,855 ($238,337/year)

Core Leader, 15%, Using Stealth Gold Nanoparticles to Improve Radiation Therapy, P01 CA257932-01A1, NIH/NCI, PI - Krishnan, 7/1/2022-6/30/2027, $1,728,851 ($348,658/year)

Principal Investigator, 15%, Translatable Molecular Ultrasound Imaging of Cancer Metastasis Using Targeted Phase-change Nanodroplets, RP220379, Cancer Prevention & Research Institute of Texas (CPRIT), 3/1/2022-2/28/2025, $1,038,677 ($346,226/year)

Co-Investigator, 10%, New Approach for Osteosarcoma: Enhancing and Monitoring NK Cell Therapy, 1P01CA275719-01, NIH/NCI, MPI – Kundra and Gopalakrishnan, 9/1/2022-8/31/2027, $20,872,005

Principal Investigator, 15%, Next-generation high-Z nanoparticle radiosensitizers to advance combination cancer therapies, 1R01CA274955-01A1, NIH/NCI, MPI – Krishnan (contact), Cho and Sokolov, 4/1/2023-3/31/2027, $2,935,950 ($755,175/year)

Co-Investigator, 5%, NK Immunotherapy for DIPG, Children's Cancer Research Fund, PI-Wintermark, 2/29/2024-1/31/2026, $249,999 ($125,000/year).

Protocols

Funded

Co-Principal Investigator, A pilot study: evaluation of polarized reflectance spectroscopy for detection of neoplasia of oral mucosa, HNS00-304, PI - Ann Gillenwater, 2002-2010

Co-Principal Investigator, Molecular Imaging of Resected Head and Neck Cancer Specimens, LAB04-0607, PI - Ann Gillenwater, 2005-2020

Co-Principal Investigator, Evaluation of Polarized Reflectance Spectroscopy for Detection of High-risk Oral Lesions, H10-01313, PI - Catherine F. Poh, 2011-2017

Unfunded

N/A

Patents and Technology Licenses

Patents

Medifor, Ltd., Surface-enhanced analytical procedures and substrates, United States, US 5837552, 6/7/1995, Issued

Rice University, Methods and Apparatus for Polarized Reflectance Spectroscopy, United States, US 20020135752, 9/26/2002, Issued

Medifor, Ltd., Collective resonances in two-dimensional arrays for colloidal metal films, United States, WO 9809153, 8/30/1996, Filed

The University of Texas, Medical and Imaging Nanoclusters, United States, US 2008-61108530, 10/26/2008, Filed

The University of Texas, Plasmonic Particles and Methods of Image-Guided Photothermal Therapy, United States, 61/298,097, 1/25/2010, Filed

The University of Texas, Pulsed Magneto-Motive Ultrasound Imaging, United States, 61/298,801, 1/27/2010, Filed

The University of Texas, Molecular Specific Photoacoustic Imaging of biological tissue for cancer diagnosis, United States, WO 2008101019, 2/4/2010, Filed

The University of Texas, Temperature Dependent Photoacoustic Imaging, United States, 61/367,196, 7/23/2010, Filed

The University of Texas, Magneto-plasmonic nanoclusters with tunable size and optical properties, United States, 61/866,804, 8/2013, Filed

The University of Texas, Hydrophobic shielding for enhanced nanoparticle stability, United States, US2013/54628, 8/13/2013, Filed

The University of Texas, Flow chamber with size separation for capture of rare cells and photoacoustic detection, United States, 61/866,858, 8/16/2013, Filed

The University of Texas System, Methods of detecting biological activity, cellular behavior and drug delivery using encapsulated polymethine aggregates, United States, US 2015-15666750, 2/8/2015, Filed

The University of Texas System, Dye aggregates-containing nanoparticles and uses thereof, US Patent Application, 18/487,550, 11/27/2019, Filed

The University of Texas System, Cancer radiosensitization by in situ formation of gold nanoparticles and/or gold nanoclusters, United States, U.S. Patent Application, 18/003,744, 6/30/2020, Filed

Imagion Biosystems, Methods and Apparatuses for the Synthesis of Drug-loaded Magnetic Liposomes United States, U.S. Patent Application, 18/116,976, 9/16/2020, Filed

Technology Licenses

Remicalm LLC, Rice University, Biospecific Contrast Agents, US 20040023415 A1 20040205

Ovna Med, The University of Texas, Fiber-optic probes and associated methods, WO 2010065827 A1 20100610

Remicalm LLC, The University of Texas, Methods and apparatus for polarized reflectance spectroscopy, US 20020135752 A1 20020926

Remicalm LLC, Rice University, Needle Biopsy Imaging System, WO 2007022196 A2 20070222

Grant Reviewer/Service on Study Sections

Ad-hoc member of the NIH Microscopic Imaging study section, NIH, Ad Hoc Member, 2005

Mail reviewer, Department of Veteran Affairs Rehabilitation Research and Development Service, Depart of Veteran Affairs, Mail reviewer, 2006

Mail reviewer, Department of Veteran Affairs Rehabilitation Research and Development Service, Department of Veteran Affairs, Mail reviewer, 2007

NIH review panel for SBIR Topic 252 "Nanotechnology Imaging and Sensing Platforms for Improved Diagnosis of Cancer", NIH, Member, 2008

The National Center for Research Resources Special Emphasis Review Panel ZRR1 BT-B (01) 1, NIH, Member, 2008

NIH/NIGMS Special Emphasis Panel ZGM1 PPBC-5 (CP) – P01 Site Visit, NIH, Member, 2009

Special Emphasis Panel ZCA1 GRB-S M1-R - NCI Centers of Cancer Nanotechnology Excellence, NIH, Member, 2010

Exceptional, Unconventional, Research Enabling Knowledge Acceleration (EUREKA) ZCA1 SRLB-2 (M1), NIH, Member, 2011

New Biomedical Frontiers at the Interface of the Life and Physical Sciences ZRG1 BST-N 51 R, NIH, Member, 2012

Terry Fox New Frontier Program Projects, Canadian Institutes of Health Research (CIHR), Member, 2013

P41 BTRC Review Panel, NIBIB ZEB1 OSR-C J1 P, Member, 2015

Career Award and Conference Grant Review, NIH, Member, 2016

Member conflict Special Emphasis Panel Review, NIH, Member, 2016

Mentored Career Development Award (K) Application Review, NIH, Member, 2016

NCI Clinical and Translational R21 and Omnibus R03 Special Emphasis Panel, NIH, Member, 2017

NIBIB Quantum Program: Technological Innovation to Solve a Major Medical or Public Health Challenge (U01), NIH, Member, 2017

Special Emphasis Panel: Integration and Validation of Emerging Technologies to Accelerate Cancer Research, NIH, Member, 2017

Imaging Probes and Contrast Agents Study Section (IPCA), NIH, Member, 2020

Special Emphasis Panel: Cancer Detection and Therapy [ZRG1 OTC-L (02)], NIH, Member, 2020

TEP-12: Small Business Innovation Research (SBIR) contract proposals, NIH, Member, 2020

NIBIB Special Emphasis Panel: Career Development (Ks) and Conference Support (R13) [ZEB1 OSR-E (J1)], NIH, Member, 10/25/2022.

NIH Special Emphasis Panel: Career Development (Ks) and Conference Support (R13) [2024/01 ZEB1 OSR-F (J1) S], Member, 11/09/2023

NIH Enabling Bioanalytical and Imaging Technologies Study Section (EBIT), Member, 02/15 - 16/2024.

PUBLICATIONS

Peer-Reviewed Original Research Articles

1. Maskevich SA, Sokolov KV, Kivach LN, Podtynchenko SG, Streckal ND, Khodorchenko PV. SERS spectra and structure of thiamin, thiamin diphosphate and pyruvate adsorbed on silver surface. Bioorgan. Khimia 16:1552-1562, 1990.
2. Gachko GA, Kivach LN, Maskevich SA, Sokolov KV, Podtynchenko SG. Spectroscopic investigation of specific interactions of Pyruvic Acid in model systems. J. Appl. Spectrosc 52:555-560, 1990.
3. Nabiev IR, Sokolov KV, Voloshin ON. Surface-enhanced Raman Spectroscopy of Biomolecules. Part III. - Determination of local destabilization regions in the double helix. J. Raman Spectrosc 21:333-336, 1990.
4. Sokolov KV, Lutsenko SV, Nabiev IR, Nie S, Yu N-T. Surface-enhanced Raman analysis of biomedical eye lens extracts. Appl. Spectrosc 45:1143-1148, 1991.
5. Sokolov KV, Khodorchenko PV, Petukhov AV, Nabiev IR, Chumanov G, Cotton TM. Contribution of short-range and classical electromagnetic mechanisms to surface-enhanced Raman scattering from several types of biomolecules adsorbed on cold-deposited island films. Appl. Spectrosc 47:515-522, 1993.
6. Sokolov KV, Byramova NE, Mochalova LV, Tuzikov AB, Shiyan SD, Bovin NV, Nabiev IR. Detection of sialic acid residues and studies of their organization in normal and tumor a- acid

glycoproteins as probed by surface-enhanced Raman spectroscopy. Appl. Spectrosc 47:535-538, 1993.
7. Oleynikov VA, Sokolov KV, Nabiev IR. Trace Analysis by Surface- Enhanced Raman Scattering with the Use of the Track Membrane Technique. J. Appl. Spectrosc 59:820-825, 1993.
8. Chumanov GD, Sokolov KV, Gregory BW, Cotton TM. Colloidal Films as New Substrates for Surface Enhanced Spectroscopies. J. Phys. Chem. 99:9466-9471, 1995.
9. Feofanov AV, Oleinikov VA, Tuzikov AB, Ianoul AI, Kryukova EYu, Sokolov KV, Bovin NV, Nabiev IR. Surface-enhanced Raman scattering spectroscopy of sialosides and their derivatives. Bioorg. Khim 22:706-716, 1996.
10. Fritzsche W, Sokolov KV, Chumanov GD, Cotton TM, Henderson E. Ultrastructural Characterization of Colloidal Metal Films for Bioanalytical Applications by SFM. J. Vac. Sci. Technol 14:1766-1769, 1996.
11. Chumanov G, Sokolov KV, Cotton TM. Unusual Extinction Spectra of Nanometer-Sized Silver Particles Arranged in Two-Dimensional Arrays. J. Phys. Chem. 100:5166-5168, 1996.
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13. Sokolov KV, Chumanov GD, Cotton TM. Enhancement of Molecular Fluorescence near the Surface of Colloidal Metal Films. Anal. Chem. 70:3898-3905, 1998.
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15. Drezek R, Sokolov KV, Utzinger U, Boiko I, Malpica A, Follen M, Richards-Kortum R. Understanding the contributions of NADH and collagen to cervical tissue fluorescence spectra: modeling, measurements, and implications. J Biomed Opt 6:385-96, 2001. PMID: 11728196.
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17. Myakov A, Nieman LT, Wicky L, Utzinger U, Richards-Kortum R, Sokolov KV. Fiber optic probe for polarized reflectance spectroscopy in vivo: design and performance. J Biomed Opt 7:388-97, 2002. PMID: 12175288.
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19. Sokolov KV, Galvan J, Myakov A, Lacy A, Lotan R, Richards-Kortum R. Realistic three-dimensional epithelial tissue phantoms for biomedical optics. J Biomed Opt 7:148-56, 2002. PMID: 11818022.
20. Lee J, Rogers JD, Descour MR, Hsu E, Aaron JS, Sokolov KV, Richards-Kortum RR. Imaging quality assessment of multi-modal miniature microscope. Optics Express 11:1436-1451, 2003.
21. Pavlova I, Sokolov KV, Drezek R, Malpica A, Follen M, Richards-Kortum R. Microanatomical and biochemical origins of normal and precancerous cervical autofluorescence using laser-scanning fluorescence confocal microscopy. Photochem Photobiol 77(5):550-5, 2003. PMID: 12812299.
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26. Nieman L, Myakov A, Aaron J, Sokolov KV. Optical sectioning using a fiber probe with an angled illumination-collection geometry: evaluation in engineered tissue phantoms. Appl Opt 43:1308-19, 2004. PMID: 15008534.
27. Sokolov KV, Nieman LT, Myakov A, Gillenwater A. Polarized reflectance spectroscopy for pre-cancer detection. Technol Cancer Res Treat 3(1):1-14 (journal cover), 2004. PMID: 14750888.
28. Sun YP, Collier T, Aaron J, Markey M, Richards-Kortum, R, Sokolov KV, Mackinnon N, MacAulay C, Coghlan L, Milbourne A, Follen M. Multispectral digital microscopy for in vivo monitoring of oral neoplasia in the hamster cheek pouch model of carcinogenesis. Optics Express 13(3):749-762, 2005.
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36. Truskett TM, Johnston KP, Maynard JA, Borwankar AU, Murthy AK, Stover RJ, Wilson BK, Dinin AK, Laber JR, Gourisankar S, Dear B, Hung J, Sokolov KV, Hardin WG, Nie GD, Schramm R, Twu A, Miller MA. Assembling nanoclusters in water for therapy or imaging. Pap Am Chem Soc 247(1) (#208-PMSE), 2014.
37. Stover RI, Murthy A, Gourisankar S, Ne G, Martinez M, Truskett T, Sokolov K, Johnston K. Biodegradable gold nanoclusters which exhibit high-NIR absorbance for biomedical imaging. Pap Am Chem Soc 247(1) (#108-COLL), 2014.
38. Stover R, Murthy A, Gourisankar S, Martinez M, Issac A, Truskett T, Sokolov K, Johnston K. Protein resistant nanoparticle surfaces designed for the assembly of biodegradable plasmonic gold nanoclusters. Pap Am Chem Soc 249(1) (#326), 2015.
39. Romero Aburto R, Mathieu K, Kulp A, Kaffes C, Vreeland E, Sokolov K. Development of Micellar Iron Oxide Nanoparticles for Early Detection of Microscopic Cancer Disease Using

Magnetic Relaxometry. 2016 NCI Alliance for Nanotechnology in Cancer, Washington DC, 10/2016.

40. Sokolov KV, Wu CH, Cook J, Zal T, Emelianov SY. Biomedical applications of magneto-plasmonic nanoclusters (Conference Presentation). Bellingham: Spie-Int Soc Optical Engineering, 2016.
41. Harris JT, Dumani DS, Cook JR, Sokolov KV, Emelianov SY, Homan KA. Identification of M1 Macrophages in Vulnerable Plaques With Folate Receptor Beta Targeted Liposomal Indocyanine Green J-Aggregates. Circulation 134(7), 2016.
42. Geoffrey PL, Aaron J, Jeffrey NM, Gillenwater AM, Emelianov SY, Sokolov K. Plasmonic ruler: From cells to detection of micrometastasis in patients. Pap Am Chem Soc 251(1) (#316), 2016.
43. Romero Aburto R, Mathieu K, Kulp A, Vreeland E, Lu Z, Bast R, Sokolov K, Hazle J. Magnetic relaxometry detection of stealth, antibody-targeted micellar iron oxide nanoparticles in-vivo. AACR Annual Meeting 2017, Washington DC, 4/2017.
44. Romero Aburto R, Mathieu K, Vreeland E, Lu Z, Bast R, Hazle J, Sokolov K. Contrast agent development and validation for in-vivo early cancer detection using magnetic relaxometry. Cancer Nanotechnology Gordon Research Conference, West Dover VT, 6/2017.
45. Harris JT, Dumani DS, Cook JR, Sokolov KV, Emelianov SY, Homan KA. Assessment of plaque vulnerability in atherosclerosis via intravascular photoacoustic imaging of targeted liposomal ICG J-aggregates. Bellingham: Spie-Int Soc Optical Engineering, 2017.
46. Aburto RR, Sokolov K, Kulp AM, Vreeland EC, Lu Z, Bast RC, Hazle JD, Mathieu KB. Magnetic relaxometry detection of stealth, antibody-targeted micellar iron oxide nanoparticles in-vivo. Cancer Research 77(1):doi: 10.1158/1538-7445.AM2017-887. (#887), 2017.
47. Bischoff FZ, Aburto RR, Vreeland EC, Hazle JD, Sokolov K, Minser KE, Karaulanov T, Paciotti G. Detection of HER2+tumor cells using MagSense (TM) nanoparticles: safety and sensitivity. Cancer Research 78(13):2 (#060), 2018.
48. Mitcham T, Nevozhay D, Lai S, Sokolov K, Bouchard R. Ultrasound Activatable Nanodroplets Targeted to EGFR for Molecular Imaging with High-Frequency Ultrasound. Med Phys 45(6):E569-E, 2018.
49. Mitcham T, Nevozhay D, Lai S, Sokolov K, Bouchard R. Nonlinear Ultrasound Imaging of Phase-Change Perfluorocarbon Nanodroplets Activated by High-Frequency Ultrasound in a Tissue-Mimicking Environment. Med Phys 46(6):E468-E, 2019.
50. Wood C, Kim C, Han S, Wen Y, Naser M, Kaffes C, Cook J, Sood A, Sokolov K, Bouchard R. Quantitative Photoacoustic Imaging for the Assessment of Targeted Lipo-JICG in a Preclinical Model. Med Phys 46(6):E132-E, 2019.

Book Chapters

1. Nabiev I, Sokolov KV, Manfait M. Surface Enhanced Raman Spectroscopy and its Biomedical Applications, in Biomolecular Spectroscopy. In: Biomolecular Spectroscopy. 20. Ed(s) RJH Clark and RA Hester. John Wiley & Sons, 267-338, 1993.
2. Richards-Kortum R, Drezek R, Sokolov KV, Pavlova I, Follen M. Survey of Endogenous Biological Fluorophores, in Handbook of Biomedical Fluorescence. In: Handbook of Biomedical Fluorescence. Ed(s) A Mycek and B Pogue. Marcel Dekker, Inc.: New York, 237-264, 2003.
3. Sokolov KV (corresponding author), Nida D, Descour M, Lacy A, Levy M, Hall B, Dharmawardhane S, Ellington A, Korgel B, Richards-Kortum R. Molecular Optical Imaging of Therapeutic Targets of cancer, in Genomics in Cancer Drug Development. In: Advances in Cancer Research. 96. Ed(s) Hampton and Sikora. Eds Hampton and Sikora, 299-344, 2007.
4. Kan WC, Nieman LT, Sokolov KV, Markey MK. AI in clinical decision support: applications in optical imaging for cancer diagnosis. In: Advanced Computational Intelligence Paradigms in Healthcare. Ed(s) M. Sordo. Springer-Verlag, 27-48, 2008.
5. Emelianov S, Mallidi S, Larson T, Sokolov KV. Photoacoustic imaging and therapy utilizing molecular specific plasmonic nanoparticles. In: Photoacoustic Imaging and Spectroscopy. Ed(s) L Wang. CRC Press, Taylor & Francis Group, 399-407, 2009.

6. Larson T, Travis K, Joshi P, Sokolov KV. Nanoparticles for Targeted Therapeutics and Diagnostics. In: Handbook of Biomedical Optics. Ed(s) Boas, Pitris and Ramanujam. Taylor & Francis Books, Inc., 697-721, 2011.
7. Kuroda S, Yokoyama T, Tam JO, Scott AW, MA LL, Shanker M, Jin J, Goerlich C, Willcutts D, Roth JA, Sokolov KV, Johnston KP, Ramesh R. Multifunctional Tumor-Targeted Nanoparticles for Lung Cancer. In: Pulmonary Nanomedicine: Diagnostics, Imaging, and Therapeutics. Ed(s) Vij N. Pan Stanford Publishing Pte Ltd, 15-44, 2013.
8. Nevozhay D, Bouchard R, Sokolov KV. Photoacoustic Imaging with Plasmonic Nanoparticles. In: Plasmonics in Diagnostics and Therapy. In: World Scientific Publishing Company (WSPC). Ed(s) Liz-Marzan, L., 325-370, 2021.

Books (edited and written)

N/A

Letters to the Editor

N/A

Manuals, Teaching Aids, Other Teaching Publications

N/A

Other Publications

N/A

EDITORIAL AND REVIEW ACTIVITIES

Editor/Service on Editorial Board(s)

Guest Editor, Journal of Biophotonics, Special issue on Topical Problems in Biophotonics, 2012

Senior Editor, Journal of Interdisciplinary Nanomedicine (JOIN), British Society for Nanomedicine, Wiley, 2015-present

Guest Editor, Feature issue: Biophotonic materials and applications, Biomedical Optics Express, OSA Publishing, 2016

Member of Editorial Review Board

N/A

Journal Reviewer

Reviewer, Applied Spectroscopy, 2002-present

Reviewer, Cancer Letters, 2002-present

Reviewer, Journal of Biomedical Optics, 2002-present

Reviewer, Lasers in Medical Science, 2002-present

Reviewer, Medical Physics, 2002-present

Reviewer, Optics Express, 2002-present

Reviewer, Technology in Cancer Research & Treatment, 2002-present

Reviewer, Applied Optics, 2004-present

Reviewer, Nano Letters, 2004-present

Reviewer, Future Oncology, 2006-present

Reviewer, Nature Medicine, 2006-present

Reviewer, Optics Letters, 2007-present

Reviewer, Journal Physical Chemistry, 2008-present

Reviewer, Langmuir, 2008-present

Reviewer, Medical Oncology, 2008-present

Reviewer, ACS Nano, 2009-present

Reviewer, PNAS, 2009-present

Reviewer, Cancer Research, 2010-present

Reviewer, Future Medicine, 2010-present

Reviewer, WIREs Nanomedicine & Nanobiotechnology, 2010-present

Reviewer, PLoS ONE, 2011-present

Reviewer, Nature Photonics, 2012-present

Reviewer, Theranostics, 2012-present

Reviewer, JACS, 2013-present

Reviewer, Nature, 2013-present

Reviewer, Nature Nanotechnology, 2013-present

Reviewer, Advanced Functional Material, 2014-present

Reviewer, Chemistry of Materials, 2014-present

Reviewer, Nature Communications, 2015-present

Reviewer, Scientific Reports, 2015-present

Reviewer, Advanced Healthcare Materials, 2020-present

Reviewer, Advanced Science, 2020-present

Reviewer, Biomedical Optics Express, 2020-present

Reviewer, Small, 2022-present

Reviewer, ACS Applied Materials and Interfaces, 2022-present

Reviewer, Advanced Materials, 2022-present

Reviewer, Bioconjugate Chemistry, 2022-present

Reviewer, Science Advances, 2023-present

Reviewer, Nature Cancer, 2024-present

Other Editorial and Review Activities

N/A

TEACHING

Teaching Within Current Institution -

Formal Teaching

Courses Taught

N/A

Training Programs

Principal Investigator, NCI T32 Interdisciplinary Translational Pre/Postdoctoral Program in Cancer Nanotechnology

2015-present

Other Formal Teaching

N/A

Supervisory Teaching

Committees

Advisory Committees

Marc Stephen Ramirez, Ph.D., 8/2009-2012

Christopher MacLellan, Ph.D., 11/2012-11/2016

Sara Loupot, Ph.D., 10/2014-11/2018

Maureen Aliru, M.D./Ph.D., 4/2015-6/2020

Trevor Mitcham, Ph.D., 1/2016-4/2021

Supervisory Committees

N/A

Examining Committees

Marc Stephen Ramirez, Ph.D., 8/2012

Hannah J. Lee, Ph.D., 11/2014

Chair, Sara Loupot, Ph.D. Candidate, 11/2015

Direct Supervision

Undergraduate and Allied Health Students

Research Mentor, Rice University, Brian Lee, 9/2021-present

Research Mentor, Rice University, Rian Philip, 9/2021-12/2021

Medical Students

N/A

Graduate Students

Yasaman Barekatain, Rotational GSBS Student, 9/2017-12/2017

Postdoctoral Research Fellows

Linda Nieman, Ph.D., 9/2004-4/2005

Aristarchos Papagiannaros, Ph.D, 11/2010-12/2012

Kort Travis, Ph.D., 1/2011-3/2011

Rebeca Romero Aburto, Ph.D., 11/2014-1/2018

Ashok Pehere, Ph.D., 1/2016-12/2017

Rajarajeswari Muthusivarajan, Ph.D., 7/2018-12/2021

Aaron Schwartz-Duval, Ph.D., 2/2019-present

Sangheon Han, Ph.D., 5/2020-5/2023

Ali Haghighat Mesbahi, Ph.D., 11/2023-present

Clinical Residents and Fellows

N/A

Other Supervisory Teaching

Research Mentor, Instructor, Linda Nieman, Ph.D., Instructor, 9/2006-11/2009

Research Mentor, Kelsey Mathieu, Ph.D., Instructor, 6/2014-8/2018

Research Mentor, Chang Soo Kim, Ph.D., Research Scientist, 12/2015-12/2019

Research Mentor, Dmitry Nevozhay, Ph.D., Research Scientist, 1/2017-3/2020

Research Mentor, Ashok Pehere, Ph.D., Research Scientist, 1/2018-3/2019

Research Mentor, Pavel Tsitovich, Ph.D., Research Scientist, 8/2019-present

Research Mentor, Dmitry Nevozhay, Ph.D., Instructor, 3/2020-08/2022

Research Mentor, Ananthakrishnan Soundaram Jeevarathinam, Ph.D., Research Scientist, 04/01/2023-present

Research Mentor, Dmitry Nevozhay, Ph.D., Assistant Professor, 09/2022-present

Teaching Outside Current Institution

Formal Teaching

Courses Taught

Instructor, Introduction to Biomedical Engineering, The University of Texas at Austin, Course Number: BME 301, Course Hours: 48
Spring, 2003

Instructor, Engineering Probability, Statistics, & Random Processes, The University of Texas at Austin, Course Number: BME 335, Course Hours: 48
Spring, 2004

Co-Instructor, Functional Imaging: Principles, Approaches, Applications, The University of Texas at Austin, Course Number: BME 386
Fall, 2004

Instructor, Quantitative Engineering Physiology II, The University of Texas at Austin, Course Number: BME 365S, Course Hours: 48
Spring, 2005

Instructor, Quantitative Engineering Physiology II, The University of Texas at Austin, Course Number: BME 365S, Course Hours: 48
Spring, 2006

Instructor, Quantitative Engineering Physiology II, The University of Texas at Austin, Course Number: BME 365S, Course Hours: 48
Spring, 2007

Instructor, Quantitative Engineering Physiology II, The University of Texas at Austin, Course Number: BME 365S, Course Hours: 48
Spring, 2008

Invited lecturer, Nanomedicine in Healthcare, The University of Texas at Austin, Course Number: BME 387 J
Fall, 2009

Instructor, Quantitative Engineering Physiology II, The University of Texas at Austin, Course Number: BME 365S, Course Hours: 48
Spring, 2009

Instructor, Quantitative Engineering Physiology II, The University of Texas at Austin, Course Number: BME 365S, Course Hours: 48
Spring, 2010

Instructor, Quantitative Engineering Physiology II, The University of Texas at Austin, Course Number: BME 365S, Course Hours: 48
Spring, 2011

Instructor, Quantitative Engineering Physiology I, The University of Texas at Austin, Course Number: BME 365R, Course Hours: 48
Fall, 2012

Instructor, Quantitative Engineering Physiology II, The University of Texas at Austin, Course Number: BME 365R, Course Hours: 48
Spring, 2012

Invited Lecturer, Biomedical Micro and Nanotechnology, The University of Texas at Austin, Course Number: BME 385J
Spring, 2013

Instructor, Quantitative Engineering Physiology I, The University of Texas at Austin, Course Number: BME 365R, Course Hours: 48
Fall, 2013

Training Programs

N/A

Other Formal Teaching

Faculty Advisor, Senior Design Course Department of Biomedical Engineering, The University of Texas at Austin
2005-2006

Faculty Advisor, Senior Design Course, Department of Biomedical Engineering, The University of Texas at Austin
2006-2007

Supervisory Teaching

Committees

Advisory Committees

Advisory Committee, Department of Biomedical Engineering, The University of Texas at Austin, Dizem Arifler, Ph.D., 2/2004-8/2005

Advisory Committee, Department of Chemical Engineering, The University of Texas at Austin, Austin, Felice Shieh, Ph.D., 8/2004-10/2005

Advisory Committee, Department of Biomedical Engineering, The University of Texas at Austin, Bongsu Jung, Ph.D., 10/2005-9/2007

Advisory Committee, Department of Biomedical Engineering, The University of Texas at Austin, Jignesh Shah, Ph.D., 9/2006-10/2008

Advisory Committee, Department of Biomedical Engineering, The University of Texas at Austin, Jose R. Morones, Ph.D., 10/2006-6/2008

Advisory Committee, Department of Chemical Engineering, The University of Texas at Austin, Danielle Smith, Ph.D., 3/2007-2009

Advisory Committee, Department of Biomedical Engineering, The University of Texas at Austin, Bilal Ghosn, Ph.D., 12/2007-2009

Advisory Committee, Department of Chemical Engineering, The University of Texas at Austin, Jasmine Tam, Ph.D., 12/2007-8/2009

Advisory Committee, Department of Biomedical Engineering, The University of Texas at Austin, Nick Durr, Ph.D., 2008-8/2010

Advisory Committee, Department of Biomedical Engineering, The University of Texas at Austin, Kimberly A. Homan, Ph.D., 2008-11/2010

Advisory Committee, Department of Mechanical Engineering, The University of Texas at Austin, Christopher Hoy, Ph.D., 7/2008-2011

Advisory Committee, Department of Biomedical Engineering, The University of Texas at Austin, Srivalleesha Mallidi, Ph.D., 8/2008-8/2009

Advisory Committee, Department of Biomedical Engineering, The University of Texas at Austin, Narasimhan Rajaram, Ph.D., 8/2008-12/2009

Advisory Committee, Department of Chemical Engineering, The University of Texas at Austin, Li Leo Ma, Ph.D., 10/2008-12/2009

Advisory Committee, Department of Biomedical Engineering, The University of Texas at Austin, Jae Sook Park, Ph.D., 5/2009-2011

Advisory Committee, Department of Biomedical Engineering, The University of Texas at Austin, Doug Yeager, Ph.D., Ph.D. (Candidate), 8/2009-7/2014

Advisory Committee, Department of Biomedical Engineering, The University of Texas at Austin, Priyaveena Puvanakrishnan, Ph.D., 4/2010-2011

Advisory Committee, Department of Biomedical Engineering, The University of Texas at Austin, Prinda Wanakule, Ph.D., 6/2010-11/2012

Advisory Committee, Department of Biomedical Engineering, The University of Texas at Austin, Jimmy Su, Ph.D., 7/2010-8/2011

Advisory Committee, Department of Biomedical Engineering, The University of Texas at Austin, Seungsoo Kim, Ph.D., 9/2010-8/2011

Advisory Committee, Department of Biomedical Engineering, The University of Texas at Austin, Katheryne Wilson, Ph.D., 11/2010-4/2012

Advisory Committee, Department of Biomedical Engineering, The University of Texas at Austin, Jason Cook, Ph.D., Ph.D. (Candidate), 10/2011-4/2013

Advisory Committee, The University of Texas at Austin, Yu-Yen (Eric) Huang, Ph.D., Ph.D. (Candidate), 10/2011-12/2013

Advisory Committee, Department of Electrical Engineering, The University of Texas at Austin, Yun-Sheng Chen, Ph.D., 11/2011-7/2012

Advisory Committee, Department of Biomedical Engineering, The University of Texas at Austin, Min Qu, Ph.D., Ph.D. (Candidate), 5/2012-12/2013

Advisory Committee, Department of Biomedical Engineering, The University of Texas at Austin, Geoffrey Luke, Ph.D., Ph.D. (Candidate), 9/2012-8/2013

Advisory Committee, Department of Biomedical Engineering, The University of Texas at Austin, Xiaojia Mu, Ph.D., Ph.D. (Candidate), 9/2012-12/2013

Advisory Committee, Department of Biomedical Engineering, The University of Texas at Austin, Laura Ricles, Ph.D., Ph.D. (Candidate), 11/2012-8/2014

Advisory Committee, Department of Chemical Engineering, The University of Texas at Austin, Ameya Borwankar, Ph.D., Ph.D. (Candidate), 12/2012-12/2014

Advisory Committee, Department of Chemical Engineering, The University of Texas at Austin, Bobby Stover, Ph.D., Ph.D. (Candidate), 12/2012-9/2015

Advisory Committee, Department of Biomedical Engineering, The University of Texas at Austin, Navid Ghorashian, Ph.D., Ph.D. (Candidate), 2/2013-11/2013

Advisory Committee, Department of Biomedical Engineering, The University of Texas at Austin, Soon Joon Yoon, Ph.D., Ph.D. (Candidate), 7/2013-7/2014

Advisory Committee, Department of Chemical Engineering, The University of Texas at Austin, Ehsan Moaseri, Ph.D., Ph.D. (Candidate), 8/2016-8/2019

Advisory Committee, Department of Biomedical Engineering, The University of Texas at Austin, Chih Hao Liu, Ph.D., Ph.D. (Candidate), 8/2018-8/2019

Advisory Committee, Department of Bioengineering, Rice University, Adam Yaseen, Ph.D. (Candidate), 11/2023-present

Supervisory Committees

N/A

Examining Committees

Examining Committee, Department of Molecular Medicine, City of Hope, Los Angeles, CA, Alice Liu, Ph.D., Ph.D., 4/2018

Direct Supervision

Undergraduate and Allied Health Students

Research Mentor, The University of Texas at Austin, Stephen Mathew, 9/2004-5/2006

Research Mentor, The University of Texas at Austin, Tim Larson, 5/2005-5/2006

Research Mentor, The University of Texas at Austin, Kristin Wall, 8/2005-5/2006

Research Mentor, The University of Texas at Austin, Allison Nitsch, 9/2005-5/2007

Research Mentor, The University of Texas at Austin, Josh Laird, 1/2006-7/2006

Research Mentor, The University of Texas at Austin, Kyle Chai, 1/2006-12/2006

Research Mentor, The University of Texas at Austin, Philip Abitua, 5/2006-8/2006

Research Mentor, The University of Texas at Austin, Tiffany Vo, 2/2007-5/2007

Research Mentor, The University of Texas at Austin, Zachary White, 6/2007-8/2007

Research Mentor, The University of Texas at Austin, Marco Jakovljevic, 6/2007-5/2009

Research Mentor, The University of Texas at Austin, Andre Esteva, 1/2009-5/2009

Research Mentor, The University of Texas at Austin, Angela Hsiao, 5/2009-9/2009

Research Mentor, The University of Texas at Austin, Austin Yoo, 5/2009-12/2009

Research Mentor, The University of Texas at Austin, John Kim, 8/2009-12/2009

Research Mentor, The University of Texas at Austin, Angel Zubieta, 1/2010-5/2013

Research Mentor, The University of Texas at Austin, Neil Patodia, 6/2010-8/2012

Research Mentor, The University of Texas at Austin, Denny Nguyen, 1/2011-12/2012

Research Mentor, The University of Texas at Austin, Gary Re, 3/2011-12/2012

Research Mentor, The University of Texas at Austin, Vincent (Yuanchen) Wang, 4/2011-5/2013

Research Mentor, The University of Texas at Austin, Kevin Yan, 6/2011-5/2013

Research Mentor, The University of Texas at Austin, Ehssan Faraji, 8/2011-12/2011

Research Mentor, The University of Texas at Austin, Cody Hill, 10/2011-5/2012

Research Mentor, The University of Texas at Austin, Aziz Hammoud, 4/2012-12/2012

Research Mentor, The University of Texas at Austin, Priya Jain, 8/2012-1/2014

Research Mentor, The University of Texas at Austin, Jeff Sun, 9/2012-12/2012

Medical Students

N/A

Graduate Students

Supervisor, Department of Electrical Engineering, The University of Texas at Austin, Alexey Myakov, MS, MS, 9/1998-3/2003

Supervisor, Department of Physics, The University of Texas at Austin, Linda Nieman, PhD, PhD, 1/2001-8/2004

Supervisor, Department of Biomedical Engineering, The University of Texas at Austin, Jesse Aaron, PhD, Ph.D., 9/2002-4/2007

Supervisor, Department of Physics, The University of Texas at Austin, Kort Travis, MS, MS, 5/2003-12/2003

Supervisor, Department of Biomedical Engineering, The Univesity of Texas at Austin, Sonia Kumar, PhD, PhD, 5/2003-4/2007

Supervisor, Department of Electrical Engineering, The University of Texas at Austin, Sharmila Chockalingam, MS, MS, 1/2004-9/2005

Supervisor, Department of Physics, The University of Texas at Austin, Nathan Harrison, PhD, Ph.D ., 5/2004-11/2009

Supervisor, Department of Biomedical Engineering, The University of Texas at Austin, Kort Travis, PhD, Ph.D, 6/2006-11/2010

Supervisor, Department of Biomedical Engineering, The University of Texas at Austin, Tim Larson, PhD, PhD, 6/2006-7/2012

Supervisor, Department of Biomedical Engineering, The University of Texas at Austin, Justina Tam, PhD, PhD, 9/2006-11/2012

Co-Supervisor, Biomedical Engineering, The University of Texas at Austin, Chih-Wen Kan, Ph.D., PhD, 4/2008-8/2010

Co-Supervisor, Chemical Engineering, The University of Texas at Austin, Avinash Murthy, PhD, PhD, 1/2009-8/2013

Supervisor, Biomedical Engineering, The University of Texas at Austin, Chun-Hsien (Frank) Wu, PhD, 11/2010-7/2014

Supervisor, Chemistry, The University of Texas at Austin, Albert Lee, 1/2011-5/2015

Supervisor, Biomedical Engineering, The University of Texas at Austin, Maria Jimenez Bailey, PhD, 1/2011-2016

Supervisor, Biomedical Engineering, The University of Texas at Austin, Pratixa Joshi, PhD, 9/2011-4/2014

Co-Supervisor, Department of Biomedical Engineering, The University of Texas at Austin, Jason Haber, 1/2014-2016

Supervisor, Department of Bioengineering, Rice University, Sangheon Han, Ph.D., 1/2016-4/2020

Co-supervisor, Department of Bioengineering, Rice University, Bakai Sheyitov, 11/2023-present

Postdoctoral Research Fellows

Research Mentor, The University of Texas at Austin, Department of Biomedical Engineering, Sonia Kumar, Ph.D., Postdoctoral Fellow, 5/2007-8/2007

Research Mentor, The University of Texas at Austin, Department of Biomedical Engineering, Aaron Jesse, Ph.D., Postdoctoral Fellow, 5/2007-10/2008

Research Mentor, The University of Texas at Austin, Department of Biomedical Engineering, Veronika Sapozhnikova, Ph.D., Postoctoral Fellow, 9/2007-9/2009

Research Mentor, The University of Texas at Austin, Leonid Fradkin, Ph.D, Postdoctoral Fellow, 6/2010-2/2012

Clinical Residents and Fellows

N/A

Other Supervisory Teaching

N/A

CONFERENCES AND SYMPOSIA

Organization of Conferences/Symposia (Include chairing session)

OSA, Biomedical Optical Spectroscopy Imaging & Diagnostics (BOSD), Miami Beach, FL, Program Committee Member, 2004

International Symposium, Topical Problems of Biophotonics, Moscow, Russia, Moscow, Program Committee Member, 8/2007

OSA, Topical Meeting on Biomedical Optics, St. Petersburg, FL, Program Committee Member, 3/2008

OSA, Topical Meeting, Optical Molecular probes, Imaging, and Drug Delivery, Monterey, CA, Chair, 4/2011

III International Symposium, III International Symposium on Topical Problems of Biophotonics,, St. Petersburg, Russia, Russian Federation, Chair, 7/2011

OSA, Optics and Photonics Congress, Topical Meeting: Optical Molecular probes, Imaging, and Drug Delivery, Waikoloa Beach, HI, Chair, 4/2013

SPIE, SPIE Photonics West BiOS, Topical Meeting on Colloidal Nanoparticles for Biomedicial Applications, San Francisco, CA, Program Committee Member, 2/2015

V International Syposium, V International Syposium on Topical Problems of Biophotonics, Nizhny, Russian Federation, Co-Chair of the Topical Conference on Nanobiophotonics, 7/2015

15th International Conference on Laser Applications in Life Sciences, Faculty of Engineering, Bar-Ilan University, Tel-Aviv, Israel, Session Chair, 11/2018

Presentations at National or International Conferences

Invited

Engineering three-dimensional epithelial tissue for biomedical optics, OSA Biomedical Topical Meeting on Biomedical Optical Spectroscopy and Diagnostics, Miami Beach, FL, 4/7/2002

Optical Imaging with Nanoparticles, Advances in Imaging-Guided Diagnosis and Therapy, Round Top, TX, 4/15/2004

Molecular Imaging of Carcinogenesis with Immuno-Targeted Gold Bioconjugates, 26th Annual International Conference of the IEEE Engineering in Medicine and Biology Society, San Francisco, CA, 9/1/2004

Molecular imaging of carcinogenesis with metal nanoparticles, 88th OSA Annual Meeting: Frontiers in Optics, Rochester, NY, 10/10/2004

Gold Nanoparticles: a New Platform for Molecular Specific Screening, Diagnosis, and Elucidation of Cancer, In Vivo. 8th International Conference on Optics Within Life Sciences, Melbourne, Australia, 11/28/2004

Nanotechnology in Molecular Imaging and Sensing of Cancer, ASMEs 3rd Annual Bioengineering Technology Seminar, Irvine, CA, 4/5/2005

Cancer Imaging with Nobel Nanoparticles, PhAST: Photonic Applications Systems Technologies, Baltimore, MD, 5/23/2005

Polarization Spectroscopy and Imaging of Pre-Cancer in the Oral Cavity., FACSS: Federation for Analytical Chemistry and Spectroscopy Societies, Quebec City, Quebec, Canada, 10/10/2005

Gold Nanoparticle Platform for Molecular Specific Screening, Diagnosis and Therapy of Cancer In Vivo, NOEHS: Nanotechnology Occupational and Environmental Health and Safety Research to Practice, Cincinnati, OH, 12/4/2006

Plasmonic nanoparticle platform for carcinogenesis imaging in situ, International Conference on Topical Problems of Biophotonics, Nizhny Novgorod, Moscow, Russia, 8/4/2007

Optical systems for in vivo molecular imaging of cancer, 4th FICBAC Workshop on Nanotechnology for Cancer Diagnosis and Treatment, Madrid, Spain, 11/23/2007

Nanoparticles in Molecular Imaging of Cancer: Opportunities and Challenges, OSA Topical Meeting on Biomedical Optics, St. Petersburg, FL, 3/2008

Effect of Plasmon Resonance Coupling for Molecular Imaging and Therapy of Cancer, Southwest Chapter of the AAPM Meeting, Austin, TX, 4/18/2008

Multifunctional and Multimodal Nanoparticles for Molecular Specific Detection and Therapy of Cancer, BIT Annual World Cancer Congress, Shanghai, China, China, 6/2008

Imaging Molecular Assemblies Using Plasmon Resonance Coupling, Microscopy and Microanalysis Meeting, Albuquerque, New Mexico, New Mexico, 8/2008

Molecular Imaging of Carcinogenesis with Plasmonic Nanosensors, PASI Advanced Microscopy Techniques, Cancun, Mexico, Mexico, 8/2008

Cancer Imaging and Therapy with metal Nanoparticles, IEEE Engineering in Medicine and Biology Society, IEEE, Minneaplis, MN, 9/2009

Dynamic Molecular Imaging Using Nanoparticle Plasmon Resonance Coupling, SPIE Photonics West, SPIE, San Francisco, CA, 1/2010

Multifunctional and Multimodal Plasmonic Nanoparticles in Imaging and Therapy, Microscopy and Microanalysis Meeting, Portland, OR, 8/2010

Receptor Mediated Aggregation of Plasmonic Nanoparticles in Optical and Photoacoustic Molecular Imaging of Cancer, SPIE/OSA European Conferences on Biomedical Optics, Munich, Germany, 5/2011

Plasmonic Nanosensors for Molecular and Functional Cellular Imaging In Vivo, III International Symposium, St. Petersburg, Russia, 7/2011

Plenary lecture: Plasmon Resonance Coupling in Biomedical Applications: from Molecular Imaging in Cells to Functional Imaging In Vivo, XVI International School on Optics, Laser Physics and Biophotonics,, Saratov, Russia, 9/2012

Concept of Nanoparticle Clustering in Biomedical Applications, SPIE Photonics West, San Francisco, CA, 2/5/2013

Magneto-plasmonic nanoparticle platform for capture, separation and enumeration of rare cells, Nanomedicine, Edinburgh, Scotland, 3/2014

Plasmonic Nanosensors for Molecular and Functional Cancer Imaging and Therapy, SPIE Optics + Photonics, San Diego, CA, 8/2014

Biodegradable Plasmonic Nanoparticles: Overcoming Clinical Translation Barriers, OSA Congress, Optics in the Life Sciences, Vancouver, Canada, 4/2015

Interdisciplinary Translational Pre/Postdoctoral Program in Cancer Nanotechnology, NCI Alliance for Nanotechnology in Cancer Meeting, Bethesda, MD, 11/2015

Biomedical Applications of Magneto-Plasmonic Nanoclusters, SPIE Photonics West, San Francisco, CA, 2/13/2016

Plasmonic ruler: from cells to detection of micrometastasis in patients., American Chemical Society (ACS) National meeting, San Diego, CA, 3/14/2016

Biodegradable plasmonic nanoparticles for cancer imaging and therapy., International Congress of Hyperthermic Oncology, New Orleans, LA, 4/15/2016

Nanoparticle clusters for cancer imaging and therapy., American Society for Photobiology meeting, Tampa Bay, FL, 5/22/2016

Mechanism of molecular cancer imaging and therapy with ultra-small nanoparticles., VI International Symposium: Topical Problems of Biophotonics, St. Petersburg, Russian Federation, 7/20/2017

Photoacoustic molecular imaging of cancer metastasis using targeted gold nanoparticles: from pre-clinical studies to clinical translation., Indo-U.S. Symposium on Nanotechnology Regulatory Science, Hyderabad, India, 2/2018

Molecular imaging of cancer cells with ultra-small 5nm gold nanoparticles., 28th Annual Conference of the German Society for Cytometry., Jena, Germany, 9/19/2018

Laser applications in multiscale molecular imaging using ultra-small plasmonic nanoparticles-, 15th International Conference on Laser Applications in Life Sciences, Tel-Aviv, Israel, 11/21/2018

Interdisciplinary Translational Pre/Postdoctoral program in Cancer Nanotechnology, NCI Alliance for Nanotechnology in Cancer PI Meeting, Rockville, MD, 9/24/2019

Nanoparticle Theranostics, Cancer Theranostics and Molecular Imaging Symposium, Houston, TX, 11/21/2019

Clinically translatable molecular photoacoustic imaging: requirements and approaches, SPIE Photonics West, San Francisco, CA, 3/5/2021

In situ cancer cell specific biomineralization to overcome nanoparticle delivery barriers and sensitize pancreatic cancer to radiotherapy, NCI Alliance for Nanotechnology in Cancer PI Meeting, Virtual Meeting, 10/18/2022

Studies on Phase-change Nanodroplets for Biomedical Imaging, SPIE Photonics West, San Francisco, CA, 1/31/2023

Other, Including Scientific Exhibitions

Surface-enhanced Raman studies of membrane proteins and nucleic acids: topography and secondary structure of biomolecules, Spectroscopy of Biological Molecules. State of the Art, 1/1/1989

Determination of the local destabilization regions in the DNA double helix as probed by surface-enhanced Raman spectroscopy, the XII International Conference on Raman Spectroscopy, 1/1/1990

Surface-Enhanced Raman Spectroscopy of Cocarboxylase and Pyruvate Complexes, the XII International Conference on Raman Spectroscopy, 1/1/1990

Surface-Enhanced Raman studies of nucleic acids and their complexes with anti-tumour drugs, Structure and conformational dynamics of biomacromolecules. Europhysics Conference, 1/1/1990

Polynuclear membranes as a substrate for obtaining surface-enhanced Raman scattering films, Laser Applications in Life Sciences, 1/1/1991

Surface-Enhanced Raman spectroscopy in the structural studies of biomolecules, Laser Applications in Life Sciences, 1/1/1991

Surface-Enhanced Raman Spectroscopy of Nucleic Acids and Antitumour Drugs/Nucleic Acids Complexes in vitro and in Living Cells, Spectroscopy of Biological Molecules, 1/1/1991

Comparative study of structure and biological activity of synthetic and natural inhibitors of influenza virus as probed by SERS, the XIII International Conference on Raman Spectroscopy, 1/1/1992

Contribution of short range and classical electromagnetic mechanisms to SERS from different types of biomolecules adsorbed on cold-deposited island film, the XIII International Conference on Raman Spectroscopy, 1/1/1992

Surface-Enhanced Raman Spectroscopy of Sialosides Including Glycoproteins, the XVI International Carbohydrate Symposium, Paris, France, 1/1/1992

Intercalating and Non-Intercalating Antitumor Drugs: Structure-Function Correlations As Probed by Surface-Enhanced Raman Spectroscopy, Spectroscopy of Biological Molecules, 1/1/1993

Practical Substrates and Techniques for Analytical Testing, Identification and Structural Studies of Biological, Biomedical and Related Organic Molecules by Surface-Enhanced Raman Spectroscopy, Spectroscopy of Biological Molecules, 1/1/1993

SERS Spectroscopy - Physical Probing of Cell Surface Sialosides., 12th International Symposium on Glycoconjugates, 1/1/1993

Colloidal Metal Films as a New Substrate for Enhanced Spectroscopy of Biomolecules, Spectroscopy of Biological Molecules, 1/1/1995

Colloidal Metal Films as a New System for Enhanced Optical Phenomena, Nineteeth DOE Solar Photochemistry Research Conference, 1/1/1995

New Substrates for Enhanced Spectroscopies, Laser Techniques for Surface Science II, 1/1/1995

Development of New Substrates and Approaches to SERS-Based Bioanalytical Applications, the XVth International Conference on Raman Spectroscopy, 1/1/1996

SERS of Long Chain Alkanethiols on Roughened Metal Electrodes, the XVth International Conference on Raman Spectroscopy, 1/1/1996

Quantitative Optical Spectroscopy for Earlier Cancer Diagnosis, NASA/NCI Workshop on Sensors for Bio-Molecular Signatures, Pasadena, CA, 6/2/1999

Polarization Spectroscopy of Cervical Epithelium for Pre-Cancer Detection, UEF Conference on Advances in Optics for Biotechnology, Medicine and Surgery, Kona, HI, 8/1/1999

Measurement of Cellular and Nuclear Morphology In Vivo Using Reflectance Spectroscopy with Polarized Light, Gordon Conference on Lasers in Medicine and Biology, New London, CT, 6/10/2000

Polarized Reflectance Spectroscopy In Vivo: What Can We Measure?, UEF Conference on Advances in Optics for Biotechnology, Medicine and Surgery, Banff, Canada, 7/22/2001

Metal Nanoparticles as Biospecific contrast agents for cancer imaging, OSA Biomedical topical meeting on biomedical optical spectroscopy and diagnostics, Miami Beach, FL, 4/7/2002

Polarized Reflectance Spectroscopy Instrument for the Clinical Setting, Optical Society of America Biomedical Topical Meetings, Biomedical Optical Spectroscopy and Diagnostics, Miami, FL, 4/7/2002

Metal Nanoparticles as Contrast Agents for Molecular Imaging of Pre-Cancers, Gordon Conference on Lasers in Medicine and Biology, Kimball Union Academy, NH, 7/14/2002

Optical Sectioning of Epithelium Using Fiber Optic Probe with an Angled Illumination-Collection Geometry: Evaluation in Engineered Tissue Phantoms, Gordon Research Conference, Lasers in Medicine and Biology, Meriden, NH, 7/14/2002

Fiber Optic Probe for Polarized Reflectance Spectroscopy: Mucosa Studies, The Second Joint Conference of the IEEE Engineering in Medicine and Biology Society and the Biomedical Engineering Society, Houston, TX, 10/22/2002

Molecular Specific Contrast Agents to Enhance Optical Imaging of Pre-Cancers, Second Joint EMBS-BMES Conference, Houston, TX, 10/23/2002

Targeting the Depth of Elastically Scattered Light in Tissue Using a Fiber Probe with an Angled Illumination-Collection Geometry: Evaluation in Engineere, SPIE, Photonics West, San Jose, CA, 1/25/2003

Detecting Dysplasia in the Oral Cavity Using Polarized Reflectance Spectroscopy, Engineering Conferences International: Advances in Optics for Biotechnology, Medicine and Surgery, Banff, Canada, 4/3/2003

Correlation of microanatomical and biochemical properties of precancerous cervical tissue with its autofluorescence characteristics, The Annual Meeting of the American Association for Cancer Research, Washington, DC, 7/11/2003

Early detection of precancerous lesions in oral mucosa using polarized reflectance spectroscopy, The Annual Meeting of the American Association for Cancer Research, Washington, DC, 7/11/2003

Vital optical imaging of activity and distribution of metallo-proteases in three-dimensional organotypic cultures using gold bioconjugates, The Annual Meeting of the American Association for Cancer Research, Washington, DC, 7/11/2003

Vital optical imaging of epithelial neoplasia using topical delivery of immuno-targeted gold nano-particles, The Annual Meeting of the American Association for Cancer Research, Washington, DC, 7/11/2003

Vital Optical Imaging of Epithelial Neoplasia Using Topical Delivery of Immuno-targeted Gold Nanoparticles, Engineering Conferences International (ECI) Advances in Optics for Biotechnology, Medicine, and Surgery, Banff, Canada, 8/3/2003

Polarization spectroscopy to measure nuclear size distribution: improved pre-cancer detection, The Whitaker Foundation Annual Biomedical Engineering Research Conference, La Jolla, CA, 8/7/2003

Vital Molecular Imaging of Carcinogenesis with Gold Bioconjugates, AAPM Medical Imaging Research Symposium, San Diego, CA, 8/10/2003

"Results of a Pilot Clinical Trial for the Early Detection of Oral Mucosa Neoplasia Using Polarized Reflectance Spectroscopy," SPIE, Photonics West, San Jose, CA, 1/24/2004

"Cellular labeling of SiHa and oral biopsies with quantum dot bioconjugates," Gordon Research Conference on Lasers in Medicine and Biology, Meriden, NH, 7/4/2004

"Hyperspectral Imaging and Reflectance Spectroscopy to Elucidate the Scattering Properties of Oral Carcinogenesis," Gordon Research Conference, Lasers in Medicine and Biology, Meriden, NH, 7/4/2004

"In-Vivo Detection of Intra-Nuclear Targets Using Gold Nanoparticles - Human Papillomavirus," Gordon Research Conference, Lasers in Medicine and Biology, Meriden, NH, 7/4/2004

Cellular labeling of SiHa and oral biopsies with quantum dot bioconjugates, Gordon Research Conference on Lasers in Medicine and Biology 2004, Meriden, NH, 7/4/2004

Optical In-Vivo Detection of Intracellular Biomarkers Using Gold Nanoparticles, Society of Molecular Imaging, 4th Annual Meeting, Cologne, Germany, 9/7/2005

Optical In-Vivo Detection of Intracellular Biomarkers Using Gold Nanoparticles, BMES, Baltimore, MD, 9/28/2005

Ultrasound-based Thermal and Elasticity Imaging to Assist Photothermal Cancer Therapy-Preliminary Study, Proceeding of the 2006 IEEE Ultrasonics Symposium, 2006

Gold nanoparticles for scattering-based molecular detection of disease: Optical imaging of growth factor receptor mediated carcinogenesis, SPIE, Photonics West (BiOS), San Jose, CA, 1/21/2006

Analysis of Polarization Spectroscopy Data to Improve Imaging of Epithelial Pre-Cancers, Gordon Research Conference, Lasers in Medicine and Biology, Plymouth, NH, 7/2/2006

Increased molecular-specific image contrast with magnetically actuated gold/iron oxide hybrid nanoparticles, Gordon Research Conference, Lasers in Medicine and Biology, Plymouth, NH, 7/2/2006

Optical Detection of intra-Cellular Cancer Biomarkers, Gordon Research Conference, Lasers in Medicine and Biology, Plymouth, NH, 7/2/2006

MR Assessment of Multi-Modal Molecular Targeted Contrast Agent, ISMRM Conference, Berlin, Germany, 5/19/2007

Multifunctional Plasmonic Nanoparticles for Photothermal Therapy of Cancer with MRI Monitoring, ISMRM Conference, Berlin, Germany, 5/19/2007

Plasmon resonance coupling for sensitive monitoring of EGFR activation and trafficking in live cells, ECI Conference on Advances in Optics for Biotechnology, Medicine and Surgery, Naples, FL, 6/10/2007

Plasmonic Nanoparticles with affinity and Delivery Functionalities for Imaging Intracellular Biomarkers in Live Cells, ECI Conference on Advances in Optics for Biotechnology, Medicine and Surgery, Naples, FL, 6/10/2007

Probing the local heterogeneity of oral mucosa for the early detection of cancer using polarized reflectance spectroscopy: a pilot clinical trial, ECI Conference on Advances in Optics for Biotechnology, Medicine and Surgery, Naples, FL, 6/10/2007

Comparison of linear and non-linear classifiers for oral cancer screening by optical spectroscopy, AMIA Symposium, Chicago, IL, 11/2007

Multifaceted Approaches to Molecular Imaging Using Plasmonic Nanosensors, Material Research Society Annual Meeting, Boston, MA, 11/2007

Adaptive spectral window sizes for feature extraction from optical spectra, SPIE Photonics West, San Jose, CA, 1/2008

Molecular specific photoacoustic imaging with plasmonic nanosensors, SPIE Photonics West, San Jose, CA, 1/2008

Molecular specific photoacoustic imaging with plasmonic nanosensors, SPIE Photonics West: IX Conference on Biomedical Thermoacoustics, Optoacoustics, and Acousto-Optics, San Jose, CA, 1/2008

Phenomenology of optical scattering from plasmonic aggregates for application to biological imaging and clinical therapeutics, SPIE Photonics West, San Jose, CA, 1/2008

Photoacoustic and ultrasound imaging to guide photothermal therapy: ex vivo study, SPIE Photonics West: IX Conference on Biomedical Thermoacoustics, Optoacoustics, and Acousto-Optics,, San Jose, CA, 1/2008

Plasmon resonance coupling for monitoring membrane receptor dynamics in living cells, SPIE Photonics West, San Jose, CA, 1/2008

On possibility of molecular intravascular photoacoustic imaging using gold nanoparticles, 25th Annual Houston Conference on Biomedical Engineering Research, The Houston Society for Engineering in Medicine and Biology, Houston, TX, 2/2008

On the feasibility of in-vivo molecular intravascular photoacoustic imaging, Biomedical Engineering Society 2008 Annual Fall Meeting, St Louis, MO, 10/2008

Selective detection of cancer using spectroscopic photoacoustic imaging and bioconjugated gold nanoparticles, IEEE Ultrasonics Symposium, Beijing, China, China, 11/2008

Compact beveled multi-fiber probe for depth sensitive optical spectroscopy of precancer, SPIE, Photonics West, San Jose, CA, 1/2009

Multi-wavelength photoacoustic imaging of plasmon resonance coupling of gold nanoparticles for selective detection of cancer, SPIE Photonics West Symposium: Photons Plus Ultrasound: Imaging and Sensing, San Jose, CA, 1/2009

Optical Imaging of Interactions between Molecular-Specific Therapeutic Nanoparticles and Lung Cancer Cells: Correlation between Uptake, Distribution and Therapy, SPIE, Photonics West, San Jose, CA, 1/2009

Selective Detection of Cancer with Combined Photoacoustic and Ultrasound Imaging using Molecular Specific Gold Nanoparticles, The 2009 Central Texas Clinical Research Forum, Austin, TX, 5/2009

Contrast agents based on gold nanoparticles for non-invasive detection of cancer, Engineering Conferences International - Advances in Optics for Biotechnology, Medicine and Surgery XI, Burlington, VT, 6/2009

Optical Imaging of Interactions of Tumor-targeted, Therapeutic Gold/Iron Nanoparticles with Lung Cancer Cells: Correlation with Therapeutic Effect, Engineering Conferences International: Advances in Optics for Biotechnology, Medicine and Surgery XI, Burlington, VT, 6/2009

Simultaneous probing of multiple optical path lengths for precancer detection, Engineering Conferences International: Advances in Optics for Biotechnology, Medicine and Surgery XI, Burlington, VT, 6/2009

Molecular Ultrasound Imaging Using Magnetic Nanoprobes, IEEE EMBS Conference, Minneaplis, MN, 9/2009

On sensitivity of molecular specific photoacoustic imaging, IEEE Engineering in Medicine and Biology Society, Minneapolis, MN, 9/2009

Biodegradable Near-Infrared Plasmonic Nanoclusters for Biomedical Applicaton, SPIE Photonics West, San Francisco, CA, 1/2010

Dynamic Molecular Imaging Using Nanoparticle Plasmon Resonance Coupling, SPIE Photonics West, San Francisco, CA, 1/2010

Intravascular Photoacustic Imaging of Macrophages using Molecular Targeted Gold Nanoparticles, SPIE Photonics West, San Francisco, CA, 1/2010

Rapid Conjugation of Nucleic Acids to Gold Nanoparticles for Cancer Cell Targeting, SPIE Photonics West, San Francisco, CA, 1/2010

Synthetic Strategies for delivery, internalization, and activation of Nucleic Acids, Wyss Institute: New Directions in Synthetic Biology, Harvard Medical School, Boston, MA, 2010

Biodegradable Near-Infrared Plasmonic Nanoclusters for Biomedical Applications, Gordon Research Conference, Lasers in Medicine and Biology, Holderness, NH, 7/2010

Development of Molecular Specific Gold Nanorods for Non-invasive Imaging and Therapy, Gordon Research Conference, Lasers in Medicine and Biology, Holderness, NH, 7/2010

Biodegradable Near-Infrared Plasmonic Nanoclusters for Biomedical applications, Gordon Research Conference on Lasers in Medicine and Bioloty, Manchester, NH, 7/30/2010

Multifunctional and Multimodal Plasmonic Nanoparticles in Imaging and Therapy, Microscopy & Microanalysis 2010, Portland, OR, 8/8/2010

Imaging of Molecular Interactions in Live Cells: a Plasmonic Approach, World Molecular Imaging Congress, Kyoto, Japan, 9/2010

Molecular Imaging of Tumors In Vivo Using Nanoparticle Augmented Ultrasound Guided Photoacoustic Imaging, World Molecular Imaging Congress, Kyoto, Japan, 9/2010

Spectroscopic Intravascular Photoacoustic Imaging of Macrophages in Atherosclerotic Plaques, World Molecular Imaging Congress, Kyoto, Japan, 9/2010

Biodegradable Near-Infrared Plasmonic Nanoparticles: Clearance and Photoacoustic Imaging In Vivo, World Molecular Imaging Congress (WMIC), Kyoto, Japan, 9/11/2010

Beveled Multifiber Probes for Polarized Reflectance Spectroscopy in Tissue, BMES, Austin, TX, 10/2010

Biodegradable Near-Infrared Plasmonic Nanoclusters for Biomedical Applications, BMES, Austin, TX, 10/2010

Conjugation of Nucleic Acids to Gold Nanorods for Site-specific Delivery Using Photothermal Release, BMES, Austin, TX, 10/2010

Design of Biodegradable Gold Nanoclusters for NIR Optical Imaging, BMES, Austin, TX, 10/2010

Imaging of Molecular Interactions Between Therapeutic Gold/Iron Nanoparticles and Lung Cancer Cells, BMES, Austin, TX, 10/2010

In Vivo Tracking of Nanoparticle Labeled Mesenchymal Stem Cells Using Photoacoustic Imaging, BMES, Austin, TX, 10/2010

Microfluidic Screening Chip for Detection of Circulating Tumor Cell, BMES, Austin, TX, 10/2010

Molecular Diagnosis of Cancer using Multiplex Photoacoustic Imaging with Targeted Nanorods, IEEE International Ultrasonics Symposium, San Diego, CA, 10/2010

Molecular Specific Biocompatible Gold Nanorods for Non-invasive Imaging and Therapy, BMES, Austin, TX, 10/2010

Multifunctional Iron Oxide Core Gold Shell Nanoclusters for Optical and Magnetic Cellular Imaging, BMES, Austin, TX, 10/2010

Multiplex Photoacoustic Imaging for Molecular Imaging of Cancerous Tumors Using Targeted Nanorods, BMES, Austin, TX, 10/2010

Photoacoustic Imaging with Biodegradable Plasmonic Nanoclusters, IEEE International Ultrasonics Symposium, San Diego, CA, 10/2010

Polarization Sensitive Monte Carlo Simulation of Layered Tissues, BMES, Austin, TX, 10/2010

Targeted Delivery of Gold Coated Iron Oxide Nanoclusters for Near Infrared Cancer Imaging and MRI Contrast Enhancement, BMES, Austin, TX, 10/2010

Ultrasound Guided Photoacoustic Imaging with Gold Nanoparticles to Obtain Molecular Map of Tumor In-vivo, IEEE International Ultrasonics Symposium, San Diego, CA, 10/2010

Visualization Tools for Pol-MC to Simulate Polarized Light-Tissue Interaction, BMES, Austin, TX, 10/2010

Biodegradable NIR gold nanoclusters: photoacoustic imaging and in vivo clearance, SPIE Photonics West, San Francisco, CA, 1/2011

Model-based design of full-spectral diagnostic instrumentation, SPIE Photonics West, San Francisco, CA, 1/2011

Nanoparticle Labeling of Mesenchymal Stem Cells for In Vivo Imaging and Tracking, SPIE Photonics West, San Francisco, CA, 1/2011

Photoacoustic Imaging of Intravascular Plaques using Integrin-Targeted Gold Nanoparticles, SPIE Photonics West, San Francisco, CA, 1/2011

Thermal stability of biodegradable nanoclusters for photoacoustic imaging, SPIE Photonics West, San Francisco, CA, 1/2011

Ultrasound and Photoacoustic Imaging to Monitor Mesenchymal Stem Cells Labeled with Gold Nanoparticles, SPIE Photonics West, San Francisco, CA, 1/2011

Detection of sentinel lymph node metastasis with photoacoustic imaging, AACR, Orlando, FL, 4/2011

Design of biodegradable gold nanoclusters for NIR optical imaging, 242nd National Meeting of the American-Chemical-Society (ACS), Denver, CO, 9/2011

Combined intravascular ultrasound and spectroscopic photoacoustic imaging for detecting morphology and composition of atherosclerotic plaques, SPIE Photonics, Sam Francisco, CA, 1/2012

In vivo photoacoustic imaging of breast cancer cellular receptors using multiplex contrast agents, SPIE Photonics West, San Francisco, CA, 1/2012

Intravascular photoacoustic imaging of gold-nanorod labeled atherosclerotic plaques, SPIE Photonics West, San Francisco, CA, 1/2012

Plasmonic nanosensors for molecular photoacoustic imaging of regional micrometastasis, SPIE Photonics West, San Francisco, CA, 1/2012

Photoacoustic imaging of micrometastasis using plasmonic nanosensors, Gordon Research Conference, Lasers in Medicine and Biology,, Lasers in Medicine and Biology, Holderness,, Holderness,, NH, 7/2012

Photoacoustic Imaging of Cellular Function, Biosensing and Nanomedicine-II at SPIE Optics and Photonics, San Diego, CA, 8/2012

Detection of regional micrometastasis using plasmonic gold nanoparticles and photoacoustic imaging, BMES, Atlanta, GA, 10/2012

Applying a hydrophobic shield to diminish non-specific interactions of gold nanoparticles, SPIE Photonics West, San Francisco, CA, 2/2013

Combined SERS probes and super-resolution imaging for studying molecular interactions, SPIE Photonics West, San Francisco, CA, 2/2013

Hybrid magnetic/plasmonic nanocarriers for capture and photoacoustic detection of circulating tumor cells, SPIE Photonics West, San Francisco, CA, 2/2013

Oblique polarized reflectance spectroscopy for depth sensitive measurements in the epithelial tissue, SPIE Photonics West, San Francisco, CA, 2/2013

Charged gold nanoparticles with essentially zero serum protein adsorption in undiluted fetal bovine serum,, ACS National Meeting, New Orleans, LA, 4/2013

Equilibrium Gold Nanoclusters Quenched with Biodegradable Polymers,, ACS National Meeting, New Orleans, LA, 4/2013

Molecular photoacoustic imaging with plasmonic nanoparticles: moving toward clinical translation, OSA, Optics and Photonics Congress, Topical Meeting:, Optical Molecular probes, Imaging, and Drug Delivery, Waikoloa Beach, HI, 4/2013

Development and Optimization of Multimodal Near-IR Contrast Agents for Immune Cell Tracking, Materials Research Society, Boston, MA, 12/2013

Alternative to the sentinel lymph node biopsy: ultrasound-guided spectroscopic photoacoustic imaging of molecularly-activatable plasmonic nanosensors, SPIE Photonics West, San Francisco, CA, 2/2014

Fano-resonant mid-infrared metasurfaces: a new platform for bio-sensing and vibrational fingerprinting of proteins and cells, SPIE Photonics West, San Francisco, CA, 2/2014

Magneto-plasmonic nanoclusters for capture and photoacoustic detection of cancer cells, SPIE Photonics West, San Francisco, CA, 2/2014

Magneto-plasmonic nanoclusters with high-NIR absorbance for tracking, detection, and capture of rare cells, SPIE Photonics West, San Francisco, CA, 2/2014

Multimodal near-IR contrast agents for immune cell tracking, SPIE Photonics West, San Francisco, CA, 2/2014

Plasmonic biodegradable gold nanoclusters with high NIR-absorbance for biomedical imaging, SPIE Photonics West, San Francisco, CA, 2/2014

Depth-sensitive measurements of oral epithelial tissue with oblique polarized reflectance spectroscopy, SPIE Photonics West BiOS, San Francisco, CA, 2/2015

Enhancing immunotherapy with magneto-plasmonic nanoparticles, SPIE Photonics West BiOS, San Francisco, CA, 2/2015

Ratiometric cellular sensor based on J-aggregate trapping in liposomes, SPIE Photonics West BiOS, San Francisco, CA, 2/2015

Screening circulating tumor cells and microemboli using optical and photoacoustic microscopy, SPIE Photonics West, San Francisco, CA, 2/2015

Biodegradable Plasmonic Nanoparticles: Overcoming Clinical Translation Barriers, OSA Congress: Optics in the Life Sciences, Vancouver, Canada, 4/2015

Magneto-plasmonic nanoparticle platform for cancer diagnosis and cell based therapy, Gordon Research Conference, Cancer Nanotechnology, West Dover, VT, 6/2015

Interdisciplinary Translational Pre/Postdoctoral Program in Cancer Nanotechnology, NCI Alliance for Nanotechnology in Cancer Meeting, Bethesda, MD, 11/2015

Identification of M1 Macrophages in Vulnerable Plaques with Folate Receptor Beta Targeted Liposomal Indocyanine Green J-Aggregates., American Heart Association meeting, New Orleans, LA, 11/2016

Assessment of plaque vulnerability in atherosclerosis via intravascular photoacoustic imaging of targeted liposomal ICG J-aggregates., SPIE Photonics West BiOS, San Francisco, CA, 2/2017

Magnetic relaxometry detection of stealth, antibody-targeted micellar iron oxide nanoparticles in-vivo., AACR Annual meeting, Washington D.C., WA, 4/2017

Biodegradable nanoclusters for molecular cancer imaging and therapy., Gordon Research Conference in Cancer Nanotechnology, West Dover, VT, 6/2017

Ultra-small Plasmonic NanoSensors for Cancer Theranostics., Gordon Research Conference, Cancer Nanotechnology, West Dover, VT, 6/2017

Longitudinal elastic wave imaging using nanobomb optical coherence elastography., SPIE Photonics West, San Francisco, CA, 2/2018

Optical elastography using dye nanoparticles., SPIE Photonics West, San Francisco, CA, 2/2018

Detection of HER2+ tumor cells using MagSenseTM Nanoparticles: Safety and Sensitivity., AACR Annual meeting, Chicago, IL, 4/2018

Tissue biomechanics with nanobomb optical coherence elastography., Biomedical Optics Congress, Hollywood, FL, 4/2018

Ultrasound Activatable Nanodroplets Targeted to EGFR for Molecular Imaging with High-Frequency Ultrasound., Annual Meeting of the American-Association-of-Physicists-in-Medicine (AAPM), Nashville, TN, 7/2018

High-frequency molecular ultrasound imaging using targeted phase-change nanodroplets., IEEE Ultrasonics Symposium, 12/2018

Longitudinal elastic wave imaging using nanobomb optical coherence elastography., SPIE Photonics West, San Francisco, CA, 2/2019

Imaging of Tumor Mechanobiology Using "Nanobomb" Optical Coherence Elastography., Gordon Research Conference in Cancer Nanotechnology, West Dover, VT, 6/2019

Nonlinear Ultrasound Imaging of Phase-Change Perfluorocarbon Nanodroplets Activated by High-Frequency Ultrasound in a Tissue-Mimicking Environment., Annual Meeting of the American-Association-of-Physicists-in-Medicine (AAPM), San Antonio, TX, 7/2019

Quantitative Photoacoustic Imaging for the Assessment of Targeted Lipo-JICG in a Preclinical Model., Annual Meeting of the American-Association-of-Physicists-in-Medicine (AAPM), San Antonio, TX, 7/2019

Liposome-encapsulated ICG dye J-aggregates for molecular photoacoustic imaging of ovarian cancer, World Molecular Imaging Congress, Montreal, Canada, 9/2019

High-frequency ultrasound imaging of targeted, acoustically activated high-contrast Perfluorocarbon Nanodroplets., SPIE Medical Imaging, 2/2020

Perfluorocarbon-modified Dyes to Enhance Activatable Strategies in Cancer Imaging and Therapy, Biomedical Engineering Society Annual Meeting, San Antonio, Texas, United States, 10/2022.

Optical Coherence Elastography with Phase-Changing Perfluorocarbon Nanodroplets, Biomedical Engineering Society Annual Meeting, San Antonio, Texas, United States, 10/2022.

Prospecting Intratumoral Gold Nanoparticle Biomineralization for Cancer Therapy, Biomedical Engineering Society Annual Meeting, San Antonio, Texas, United States, 10/2022.

Comprehensive Evaluation of Molecular Specificity of ICGJ-encapsulating Agents for Clinically Translatable Molecular Photoacoustic Imaging, Biomedical Engineering Society Annual Meeting, San Antonio, Texas, United States, 10/2022.

Indocyanine Green J-Aggregate and Anticancer Drug Loaded Polymersomes for Photoacoustic Image-Guided Drug Delivery, 2022 AIChE Annual Meeting, Phoenix, AZ, United States, 11/15/2022.

Perfluorocarbon-modified Dyes to Enhance Activatable Strategies in Cancer Imaging, Gordon Research Conference in Cancer Nanotechnology, Waterville Valley, NH, United States, 06/2023.

Therapeutic disruption of the pancreatic cancer niche through in situ biosynthesis of gold nanoparticles, Gordon Research Conference in Cancer Nanotechnology, Waterville Valley, NH, United States, 06/2023.

Phase-change perfluorocarbon nanodroplets for imaging and therapy, Gordon Research Conference in Cancer Nanotechnology, Waterville Valley, NH, United States, 06/2023.

Clinically translatable molecular photoacoustic imaging agents: achieving the best molecular specificity, Gordon Research Conference in Cancer Nanotechnology, Waterville Valley, NH, United States, 06/2023.

Imaging tissue biomechanics using optical coherence elastography with blinking nanobombs, Gordon Research Conference in Cancer Nanotechnology, Waterville Valley, NH, United States, 06/2023.

Effect of Lipid Composition on Activation Potential of Phase-Shift Ultrasound Contrast Agents, AAPM 65th Annual Meeting & Exhibition, Houston, TX, United States, 07/26/2023.

Nanobomb-assisted layer-specific optical coherence elastography, Optical Elastography and Tissue Biomechanics XI, SPIE Photonics West, San Francisco, CA, United States, 02/2024.

Biodegradable polymersomes with encapsulated indocyanine green J-aggregates for highly selective photoacoustic cancer imaging, Photons Plus Ultrasound: Imaging and Sensing, SPIE Photonics West, San Francisco, CA, United States, 02/2024.

Seminar Invitations from Other Institutions

High resolution optical imaging of carcinogenesis, Smithville Science Park, Smithville, TX, 2/21/2003

Polarized Reflectance Spectroscopy of Pre-Cancer Detection, Department of Information Technology and Electrical Engineering, University of Queensland, Brisbane, Australia, 12/2/2004

Optical Imaging of Cancer with Metal Nanoparticles, Department of Physics, University of Queensland, Brisbane, Australia, 12/3/2004

Nanoparticles for Cancer Imaging, Vitesse Interdisciplinary Study Program: Biology, Vancouver, Canada, 2/24/2005

Optically active contrast agents to monitor receptor expression and interactions, Rice University Conference on Translational Optical Molecular Imaging: Nano to Macro, Rice University, Houston, TX, 5/21/2007

Imaging of Molecular Assemblies for Cancer Detection, Monitoring and Therapy: A Plasmonic Approach, Lester Wolfe Workshop in Laser Biomedicine, G R Harrison Spectroscopy Laboratory, MIT, Harvard-MIT Division of Health Science and Technology, Boston, MA, 11/25/2008

Plasmonic Nanosensors for Molecular and Functional Cancer Imaging and Therapy: Opportunities and Challenges, Texas A&M University, Biomedical Engineering, College Station, TX, 11/18/2013

Clinical translation of plasmonic nanosensors: design, synthesis and applications, Colloquia Seminar of the Chemistry Department Boston University, Boston, MA, 3/2/2015

Molecular Photoacoustic Imaging of Micrometastasis Using Plasmonic Nanosensors, Laser Microbeam and Medical Program (LAMMP) Seminar, Beckman Laser Institute and Medical Clinic, University of California, Irvine, CA, 4/8/2015

Plasmon resonance coupling: from concept to clinical translation, University of Memphis, Department of Chemistry, Memphis, TN, 1/29/2016

Molecular Photoacoustic Cancer Imaging, City of Hope, Department of Molecular Medicine, Los Angeles, CA, 4/20/2018

Enabling Nanotechnology Platforms for Cancer Imaging, BMEN Tech Talk., Texas A&M University, Department of Biomedical Engineering, College Station, TX, 2/18/2020

Clinically Translatable Nanotechnology for Cancer Molecular Imaging and Therapy, Transdisciplinary Research Webinar, Rice University, Department of Biomedical Imaging, Houston, TX, 7/29/2020

Cancer Nanotech Training Program, Multidisciplinary Training Program in Cancer Bioengineering, Rice University, Houston, TX, 10/11/2021

How a geological phenomenon can lead to development of new cancer therapies, Seminar, Department of Biomedical Engineering, University of Houston, Houston, TX, 09/16/2022

Molecular photoacoustic imaging: a winding road towards clinical translation, Departmental Seminar, Department of Biomedical Engineering, University of Houston, Houston, TX, 10/06/2023

Lectureships and Visiting Professorships

N/A

Other Presentations at State and Local Conferences

IGERT Retreat: Molecular Imaging of Carcinogenesis with Immuno-Targeted Nanoparticles, UT M.D. Anderson Cancer Center, Houston, TX, 2/11/2004

IGERT Group Meeting: Optical Spectroscopy and Molecular Imaging of Pre-Cancer, The University of Texas, Austin, TX, 8/6/2004

Molecular Imaging of Cancer with Gold Bioconjugates, Molecular Medicine Symposium, The University of Texas, Houston, TX, 2/21/2005

VaNTH Lecture: Nanoparticles for Cancer Imaging, The University of Texas, Austin, TX, 4/14/2005

Live Imaging of Epidermal Growth Factor Receptor: New Approaches and Perspectives, Grand Rounds, UT M. D. Anderson Cancer Center, Houston, TX, 7/6/2006

Nanoparticle Platform for Molecular Imaging and Therapy of Cancer, Grand Rounds, UT M. D. Anderson Cancer Center, Houston, TX, 2/16/2007

Multifunctional Plasmonic Nanosensors for Molecular Imaging and Therapy Cancer, IGERT Lecture, The University of Texas at Austin, Austin, TX, 11/2007

Nanotechnology in Medicine: Promise and Reality, Explore UT Lecture, The University of Texas at Austin, Austin, TX, 3/2008

Plasmonic Nanoparticles in Biological Imaging; Synthesis, Conjugation, Imaging and Delivery, Biomedical Optics Graduate Organization (BOGO) Seminar Series, 8/13/2008

Plasmonic Nanoparticles for Molecular Specific Imaging and Therapy of Cancer, W.M. Keck Seminar, The UT M D Anderson Cancer Center, Houston, TX, 10/13/2008

Optical Imaging, The University of Texas Biomedical Engineering Symposium, Houston, TX, 1/15/2009

Nanotechnology in Medicine: the Story of Tiny Particles, Explore UT Lecture, The University of Texas at Austin, Austin, TX, 3/2009

Nanotechnology in Medicine: The Story of Gold Particles, Health and Safety Conference on Nanotechnology, Austin, TX, 4/2009

Multifunctional and Multimodal Nanoparticles for Molecular Specific Imaging and Therapy of Cancer, BME External Advisory Committee meeting, The University of Texas at Austin, Biomedical Engineering, Austin, TX, 4/16/2010

Oblique Polarized Reflectance Spectroscopy for Cancer Detection in Epithelial Tissue, BioMedOpTex Symposium, Texas A&M University, College Station, TX, 5/25/2012

Plasmonic Nanosensors for Molecular Imaging of Cancer Micrometastasis, NorTex Nano Summit, Rice University, Houston, TX, 10/14/2014

Point of Care Molecular Cancer Imaging, MD Anderson, Image Guided cancer Therapy Research Program Seminar, Houston, TX, 9/30/2020

Culture of Safety in Research, Division of Diagnostic Imaging, MD Anderson, Houston, TX, 11/18/2021.

Imaging solutions for monitoring NK cell brain cancer immunotherapy, UT Austin and MD Anderson Collaborative Research Summit, Austin, TX, 11/02/2022

PROFESSIONAL MEMBERSHIPS/ACTIVITIES

Professional Society Activities, with Offices Held

National and International

Optical Society of America (OSA)
Member, 2007-present
Fellow, 2015-present

The Molecular Probes and Nanobio-optics Technical Group of the Optical Society of America (OSA)
Chair, 9/2008-2011

The Society of Photo-Optical Instrumentation Engineers (SPIE)
Member, 2010-present
Senior Member, 2017-present
Fellow, 2018-present

Academy of Radiology and Biomedical Imaging Research,
Distinguished Investigator, 2020-present

Local/State

N/A

UNIQUE ACTIVITIES

N/A

DATE OF LAST CV UPDATE

06/04/2024